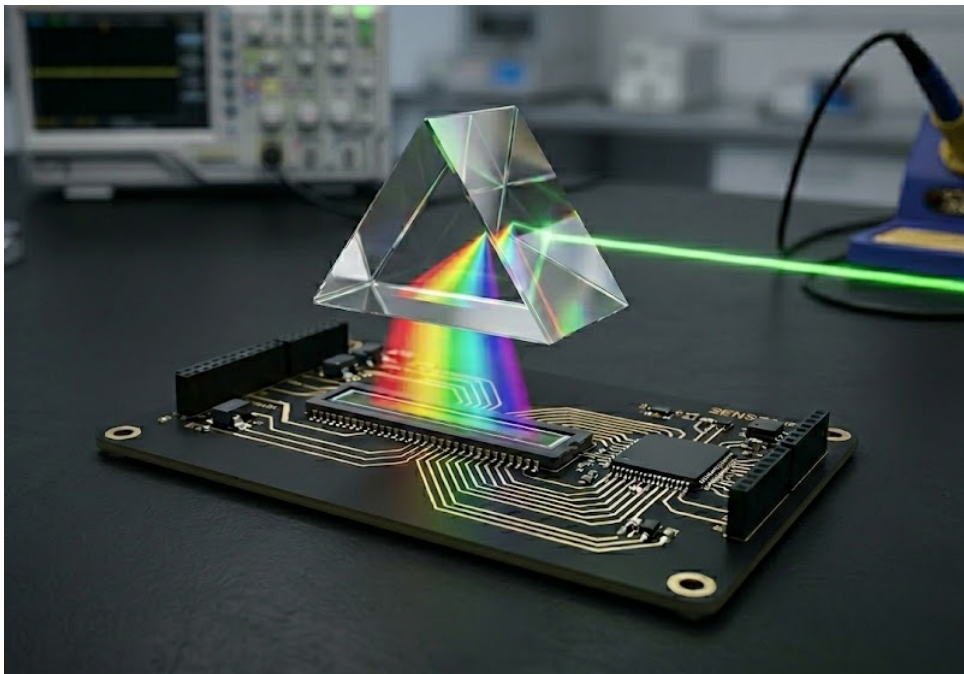


SCUOLA UNIVERSITARIA PROFESSIONALE
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Raman Spectrometer

Report



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August 20, 2026

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1. Introduction

Raman spectroscopy is a non-destructive analytical technique widely used for molecular identification and the study of materials based on their vibrational modes. The objective of this project was to develop a prototype of a portable and low-cost Raman spectrometer, jointly addressing the optical, mechanical, electronic, and software design required to acquire and analyze spectra.

The developed system uses a Toshiba TCD1304 linear CCD sensor as the detector. Its operation requires the generation of the f_M , SH, and ICG control signals, as well as the synchronized acquisition of the analog signal through the ADC of a microcontroller. During the prototyping stage, an STM32F469I-DISCO board was used, and dedicated electronics based on the STM32F446RCT6 were subsequently designed.

The architecture is completed by a Raspberry Pi responsible for communication with the microcontroller, spectrum visualization, and high-level processing. A graphical interface was developed on this platform that allows the acquisition to be configured, a dark reference to be captured, different processing tools to be applied, an interactive wavelength calibration to be performed, and the original data to be stored for later analysis.

In parallel, the optical path was designed and built using commercial components combined with mounts available in the laboratory and parts manufactured by 3D printing. The system includes the laser source, excitation and collection optics, dichroic mirror, spectral filter, slit, collimating and imaging lenses, diffraction grating, and linear detector.

The development made it possible to validate different blocks of the instrument independently, including the generation of the CCD timing signals, data acquisition and transmission, response to optical sources, resolution of lines from a neon lamp, and experimental calibration of the spectral axis. However, in the tested configurations it was not possible to reproducibly identify a Raman signal originating from the samples used.

For this reason, in addition to documenting the design and implementation developed, the report analyzes the main limitations observed during the experimental stage and proposes possible causes for the absence of Raman signal, with special attention to collection efficiency and the reproducibility of the optomechanical alignment.

The document is organized as follows:

- The **Optics and 3D design** section presents the selected components and the parts designed for their mechanical integration.
- **Acquisition, control, and processing** describes the microcontroller selection, ADC configuration, timer architecture, and the firmware responsible for controlling the CCD and transmitting the data.
- The **Hardware** section documents the design of the electronic boards, signal and power conditioning, and the auxiliary subsystems considered.
- **Signal processing** presents the fundamentals and the tools implemented for spectrum processing.
- **Acquisition and processing software on Raspberry Pi** describes the application developed for control, visualization, calibration, and data storage.
- Finally, **Raman measurement results and limitation analysis** presents the experimental tests performed and discusses the possible causes that limited Raman signal detection.

1.1 Spectral regions of interest

In Raman spectroscopy applied to biological samples, two regions of particular interest are usually considered. The so-called *fingerprint region*, approximately between 400 and 1800 cm^{-1} , contains numerous bands associated with characteristic vibrational modes of biomolecules, while the high-wavenumber region, approximately between 2800 and 3200 cm^{-1} , mainly includes C–H stretching modes.

Among the bands reported in biological samples, contributions associated with DNA, Amide III and Amide I modes, CH_2 deformations, and C=C and C–H stretching can be found [1].

These regions were used as a reference when defining the target spectral range of the instrument.

The Raman shift and the detected wavelength are related by:

$$\Delta\tilde{\nu} = \left(\frac{1}{\lambda_0} - \frac{1}{\lambda_s} \right) 10^7$$

where λ_0 is the laser wavelength and λ_s is the wavelength of the Stokes signal, both expressed in nanometers.

1.2 Excitation wavelength selection

The excitation laser wavelength represents a trade-off between Raman intensity, fluorescence, and the spectral response of the system.

Raman intensity decreases strongly as the excitation wavelength increases, so visible sources, such as 532 nm, make it possible to obtain a relatively intense signal. However, shorter wavelengths have a higher probability of exciting fluorescence in certain samples.

Near-infrared excitation is also commonly used in biological applications, particularly around 785 nm, since it can significantly reduce background fluorescence. As a counterpart, the Raman signal is weaker and may require longer integration times or higher detector sensitivity.

For this project, a 532 nm source was initially selected, prioritizing signal intensity, availability, and cost. A 785 nm source was also considered as an alternative for samples in which fluorescence could limit the measurement.

1.3 Repository

All project material, including firmware, software, 3D parts, and PCBs, is available in the public Github repository.

2. Optics and 3D design

After investigating the different possible configurations and taking the OpenRaman project as a basis, together with cost, application range, and material availability, it was decided to use:

- Laser: Edmund Optics 37-027 (5 mW, 532 nm DPSS Pointing Laser)
- Sample lens: Thorlabs AC127-019-A (f=19.0 mm, d=0.5", achromatic doublet, ARC: 400-700 nm)
- Dichroic mirror: Thorlabs DMLP550 (d=1" dichroic mirror, cut-on wavelength: 550 nm)
- Filter: Thorlabs FELH0550 (d=25 mm longpass filter, cut-on wavelength: 550 nm)
- Slit lens: Thorlabs AC127-019-A (f=19.0 mm, d=0.5", achromatic doublet, ARC: 400-700 nm)
- Slit: Thorlabs S50K (d=1" mounted slit, 50 +- 3 μ m wide, 3mm long)
- Collimating lens: Thorlabs AC254-050-A (f=50.0 mm, d=1", achromatic doublet, ARC: 400-700 nm)
- Diffraction grating: Thorlabs GR25-1205 (ruled reflective diffraction grating, 1200/mm, 500 nm Blaze, 25x25x6 mm)
- Imaging lens: Thorlabs AC254-050-A (f=50.0 mm, d=1", achromatic doublet, ARC: 400-700 nm)

2.1 3D mounts and adapters

The distribution of the elements available in the laboratory was carried out based on:

- Internal diameter of the mount
- Availability of M3 holes for distributing the Post Holders
- Extra functionality (for example, fine adjustment for the dichroic)

Quantity	Edmund Optics PN	Used for...
2	85-632	Sample lens, laser
1	85-663	Dichroic
4	87-419	Slit lens, slit, detector
2	85-630	Collimating lens, imaging lens
1	66-434	Extender (imaging system)
3	58-971, 58-951	Post holders 38 mm
1	58-970, 58-950	Post holder 25 mm
6	85-488	Rods 100 mm

In keeping with the objective of maintaining a low-cost project, 3D parts were designed to adapt the Thorlabs optical components to the Edmund Optics mounts available

in the laboratory. Additionally, auxiliary parts were developed for positioning sources and samples, reproducing optical distances, and limiting light during the alignment and calibration stages.

It was established that:

1. The outer diameter of the adapters must be 0.4 mm smaller than the inner diameters of the mounts in which they are to be placed
2. For adapters requiring a stop, the stop must have an outer diameter 0.6 mm larger than the inner diameter of the adapter
3. Areas should be opened where possible in order to save printing material

The designed adapters are detailed below.

2.1.1 Adapter for 25.4 mm lenses

Two Thorlabs AC254-050-A lenses are used: one as the collimating lens and the other as the imaging lens. Both are placed in the Edmund Optics 85-630 mount.

These mounts have an inner diameter of 30 mm, a length of 10 mm, and 3 mm holes for lens adjustment. The lenses, in turn, have a diameter of 25.4 mm and a total length of 11.5 mm, of which 2.5 mm correspond to the flat face.

To complete this assembly, the adapter was designed with an outer diameter of 29.6 mm (0.4 mm smaller than the mount) and an inner diameter of 25.8 mm (0.4 mm larger than the lens diameter). At one end of the adapter, where the flat part of the lens rests, the diameter is reduced to 23 mm. The total length is 15.5 mm, of which 2 mm correspond to this shoulder. This adapter has three 3.4 mm diameter holes, located at a height that correctly matches the 3 mm holes in the mount.

For this assembly, a 'C'-shaped stop was also designed, with an outer diameter of 26.4 mm, an inner diameter of 23 mm, an opening angle of 20° , and a length of 1.5 mm, leaving 0.5 mm free in the adapter when considering the lens + stop assembly.

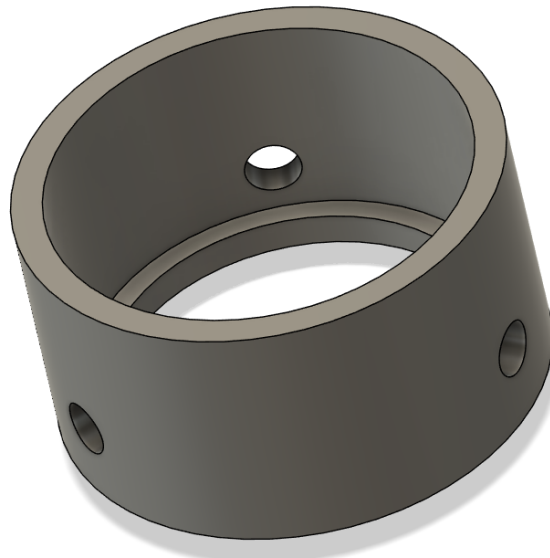


Figure 1: Adapter for 25.4 mm lenses.

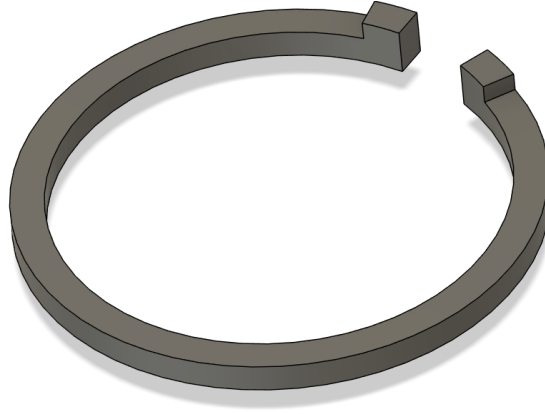


Figure 2: Example of a stop for the 25.4 mm case.

2.1.2 Adapter for 12.7 mm lenses

Two Thorlabs AC127-019-A lenses are used: one in the stage between the dichroic and the sample, and another between the filter and the slit. The first is placed in an Edmund Optics 85-632 mount, which can be placed on a Post Holder, and the other in an 87-419, supported only by the rods.

The lenses used have a diameter of 12.7 mm and a total length of 6 mm, of which 1.5 mm correspond to the flat face.

In the case of the 87-419 mount, it has an inner diameter of 25.4 mm, C-mount, a length of 10 mm, and no holes for lens adjustment. The designed adapter has an outer diameter of 25 mm, with one chamfered end down to 22 mm so that it can be inserted more easily into the C-mount. The inner diameter is 13.1 mm with a shoulder at one end that closes down to 11.4 mm. The total length is 10 mm, of which 2 mm correspond to the stop and 8 mm to the lens space. It has no adjustment holes.

In the 85-632 mount, the inner diameter is 25 mm, the length is 10 mm, and it has three 3 mm lens adjustment holes. The only differences from the previous case are that the outer diameter of the adapter is 24.6 mm, with no need for a chamfer, and that it has three 3.4 mm lens adjustment holes. The remaining parameters remain unchanged.

For both adapters, a C-shaped stop is added with an outer diameter of 13.7 mm, an inner diameter of 11.4 mm, a 20° opening, and a length of 1.5 mm, leaving 0.5 mm free in the adapter.

2.1.3 Slit adapter

A Thorlabs S50k slit is used, with a diameter of 25.4 mm, of which 9.6 mm correspond to the central circle containing the slit, and a length of 2.5 mm. It is placed in an 87-419 mount, like those already described above.

This is one of the cases in which no additional mounts of adequate diameter were available, so an adapter was designed to hold the slit from the outside. The outer diameter of the part that goes inside the mount is 25 mm, with a 22 mm chamfer for the C-mount, and the outer diameter of the part that holds the slit is 30 mm, with an inner diameter of 25.8 mm. As a shoulder, the diameter closes down to 17.2 mm and continues in this way inside the mount. The total length is 16 mm, of which 10 mm go inside the mount, 1 mm corresponds to the shoulder, and 5 mm remain available for placing the slit.

For this case, the C-shaped stop has an outer diameter of 26.4 mm, an inner diameter

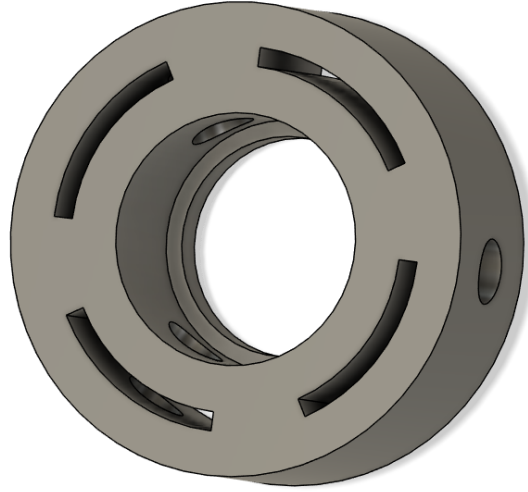


Figure 3: Adapter for 12.7 mm lenses.

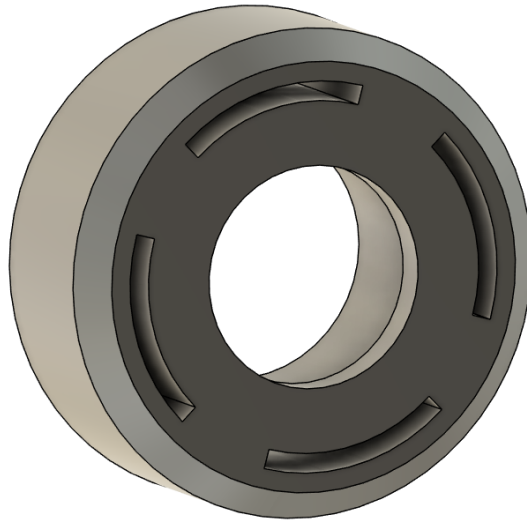


Figure 4: Adapter for 12.7 mm lenses with chamfer.

of 22.4 mm, an opening angle of 20° , and a length of 2 mm, thus leaving 0.5 mm free.

2.1.4 Filter mount

As in the previous case, a mount of adequate diameter was not available. Since the filter, although it must be correctly positioned, is not one of the most sensitive nodes in the system, it was decided to directly design the complete mount rather than only an adapter.

The Thorlabs FELH0550 was used as the filter; it has a diameter of 25 mm, a length of 3.5 mm, and an optical aperture of at least 21.1 mm. The designed mount has an outer diameter of 49.25 mm, like the Edmund Optics mounts used, an inner diameter of 25.4 mm with a shoulder that closes down to 22 mm. The total length is 10 mm, of which 3.5 mm correspond to the shoulder and 6.5 mm remain available for the filter.

It has no adjustment holes. It has four holes located 38 mm from the center for the 6.4 mm diameter rods, each accompanied by a 3.4 mm hole to secure the rod.

For the stop, a C-shape was designed with an outer diameter of 25.8 mm, an inner

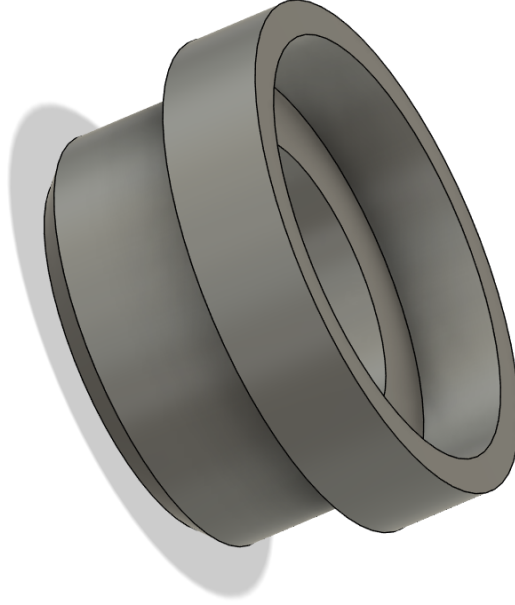


Figure 5: Slit adapter.

diameter of 22 mm, a 20° opening, and a length of 2 mm, leaving 1 mm free.

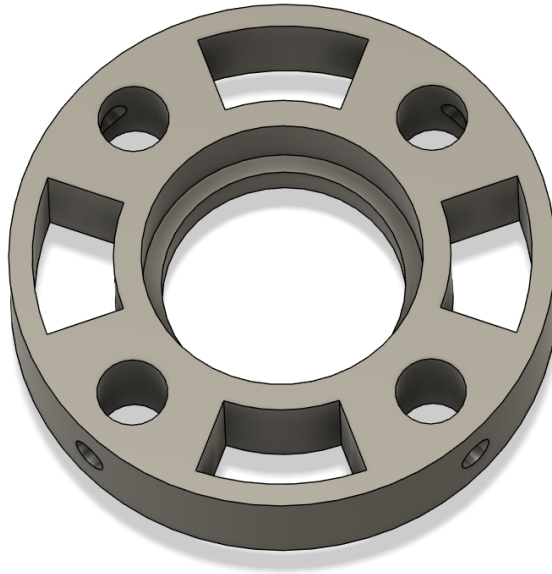


Figure 6: Filter mount.

2.1.5 Diffraction grating mount

For the grating, a complete mount also had to be designed, since its size did not allow the rod holes to be used directly, which were essential due to the limited number of mounts that accepted Post Holders.

The grating used is a Thorlabs GR25-1205, with dimensions of $25\text{ mm} \times 25\text{ mm} \times 6\text{ mm}$. The mount has an outer diameter of 49.25 mm and a space for the grating of $25.4\text{ mm} \times 25.4\text{ mm} \times 6.4\text{ mm}$. The total length is 18 mm in order to position the Post Holder at the center of the optical axis, reducing to 9 mm at the top.

Two 6.4 mm housings were added for the rods arriving from the front and another two from the side, so that the latter enter at an angle equal to the angle of incidence. The diameter for the M6 of the Post Holder is 6.4 mm and it has a depth of 7.4 mm.

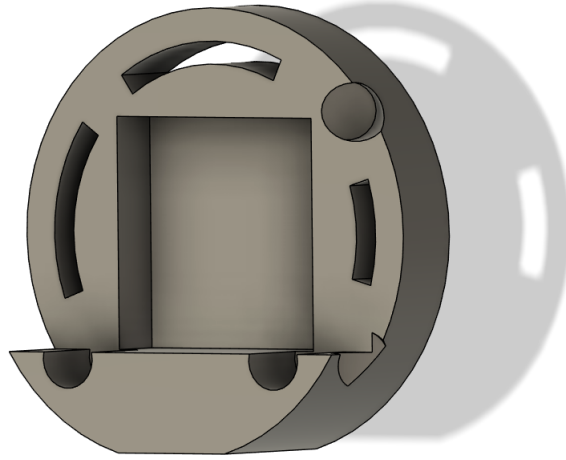


Figure 7: Diffraction grating mount.

2.1.6 Dichroic mirror adapter

The dichroic mirror case was one of the most challenging, since it was necessary not only to adapt the size but also the angle at which the mirror was placed, avoiding interruption of the rod system with an element mounted obliquely.

The mirror used is the Thorlabs DMLP550, with a diameter of 25.4 mm and a length of 3 mm. The selected mount is the 85-663, since it allows fine adjustment and has an inner diameter of 30 mm, a movable-part length of 10 mm, and 3 mm adjustment holes.

The adapter was designed with an outer diameter of 29.6 mm, an inner diameter of 25.8 mm for the region where the mirror is placed, closing onto a 23 mm shoulder. The length of the region where the mirror is located varies between 1.4 mm and 4.8 mm. The length available for insertion into the mount is 11 mm, and the adjustment holes are 3.4 mm.

Regarding the stop, the outer diameter is 26.4 mm, the inner diameter is 23 mm, the opening is 111.4° , and the length is 1.4 mm.

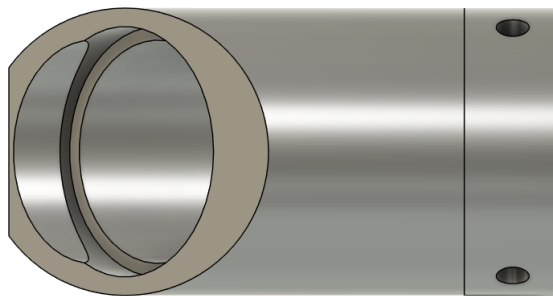


Figure 8: 45° adapter for the dichroic mirror.

2.1.7 Laser adapter

An Edmund Optics 37-027 laser available in the laboratory, with a diameter of 15 mm, is used together with an 85-632 mount, like the one described for the lens located between the sample and the dichroic.

The adapter has an outer diameter of 24.6 mm with a chamfer down to 21.6 mm in diameter, an inner diameter of 15.4 mm, and 3.4 mm adjustment holes.

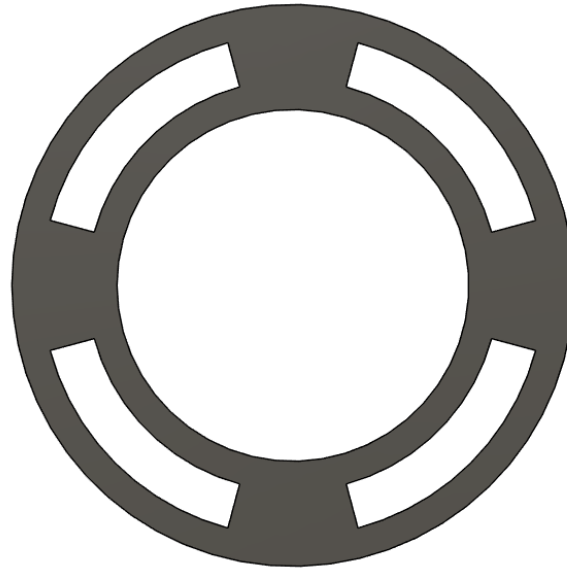


Figure 9: Laser adapter.

2.1.8 Neon lamp mount

The neon lamp was used as a reference source during alignment and spectral calibration of the instrument. In order to maintain a reproducible position with respect to the optical axis, a specific mount compatible with the rod system used in the prototype was designed.

The mount allows the lamp to be positioned either horizontally, resting on the shaped support, or vertically in the central opening, depending on the required configuration. The assembly can then be secured through the outer holes. Figure 10 shows the front and rear views of the mount.

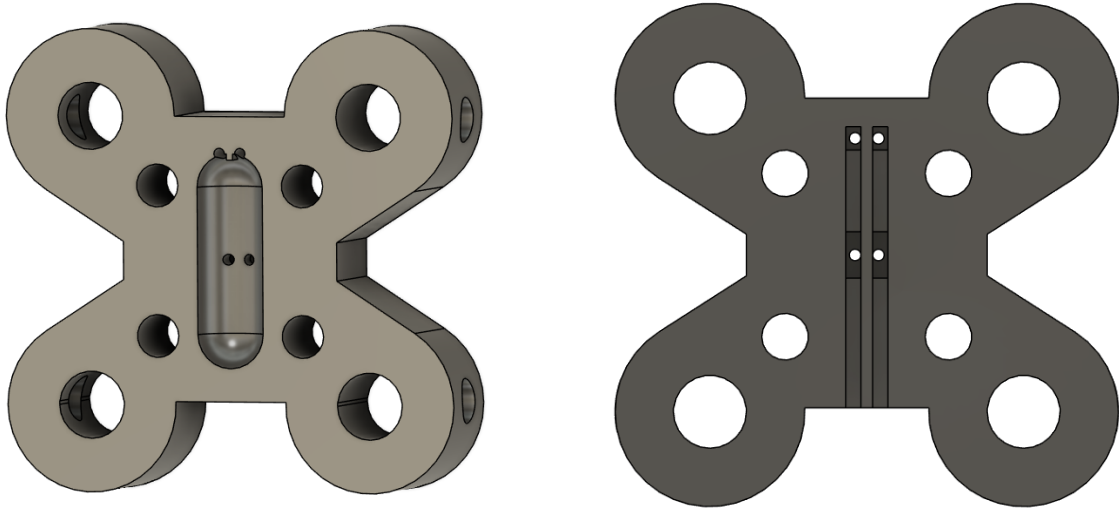


Figure 10: Mount designed for the neon lamp. Front and rear views.

2.1.9 Sample holder

A sample holder consisting of two interlocking parts was designed to position the samples in front of the excitation and collection optics. This configuration allows the same support to be adapted to different sample geometries.

The design includes two housings with different geometries, allowing the use of both a sample with an approximately square cross-section and a cuvette with a shorter optical path. In this way, the holder could be adapted to the different containers available during testing.

Figure 11 shows the parts that make up the support and a top view of the assembly.

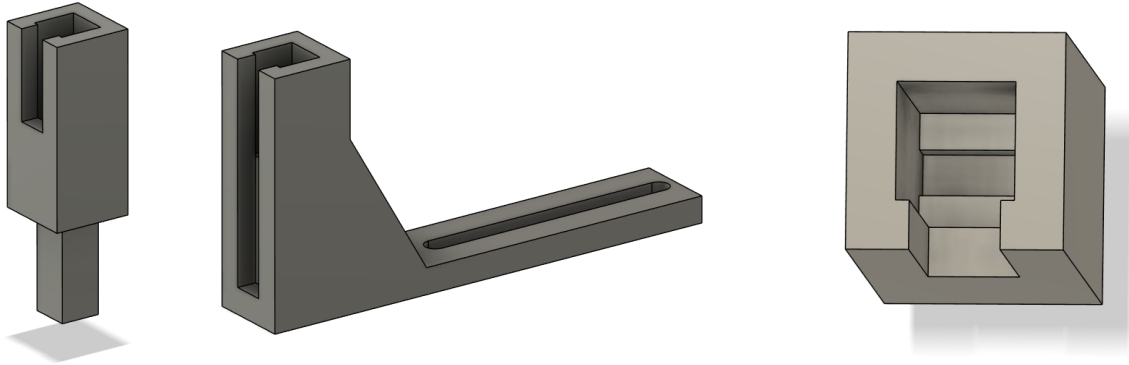


Figure 11: Sample holder designed by 3D printing. The two interlocking parts and a top view of the available geometries for positioning the sample are shown.

2.1.10 Spacers for distance adjustment

During system alignment, it was observed that small variations in the distances between optical elements could considerably modify the quality of the image obtained at the detector. Since the mounts used do not allow micrometric axial positioning, 1, 5, and 10 mm 3D-printed spacers were designed.

These parts have an open cylindrical geometry that allows them to be placed between already mounted elements without completely disassembling the assembly. Their main

function was to facilitate discrete distance adjustments and, once a satisfactory configuration had been obtained, to provide a mechanical reference for attempting to reproduce it later.

Figure 12 shows the 10 mm spacer as an example.

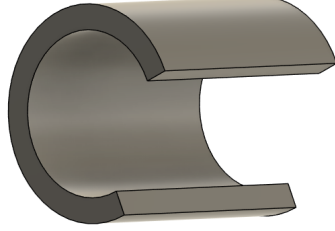


Figure 12: 10 mm 3D-printed spacer used during adjustment of the distances between optical elements.

2.1.11 Diaphragm for the neon lamp

During tests with the neon lamp, it was observed that the received intensity could be excessive and cause detector saturation. In addition, due to the mechanical implementation of the prototype, part of the radiation could propagate outside the desired optical path and reach later stages without properly passing through the slit.

To reduce both effects, a mask was designed that limits the illumination to a circular aperture with a radius of 1.5 mm. In this way, the total amount of light introduced into the system is reduced and the illuminated region is spatially restricted during alignment and calibration tasks.

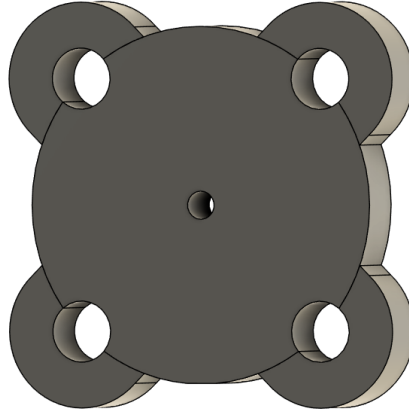


Figure 13: 3D-printed diaphragm for limiting neon lamp illumination to a circular aperture with a radius of 1.5 mm.

The need to incorporate this part also highlighted a limitation of the optomechanical assembly: the absence of an enclosure and internal diaphragms allows stray-light paths that do not exclusively follow the defined optical path. This effect is considered later in the analysis of the experimental limitations of the prototype.

2.2 Final optomechanical assembly

The complete optomechanical assembly built during the project is shown in Figure 14.

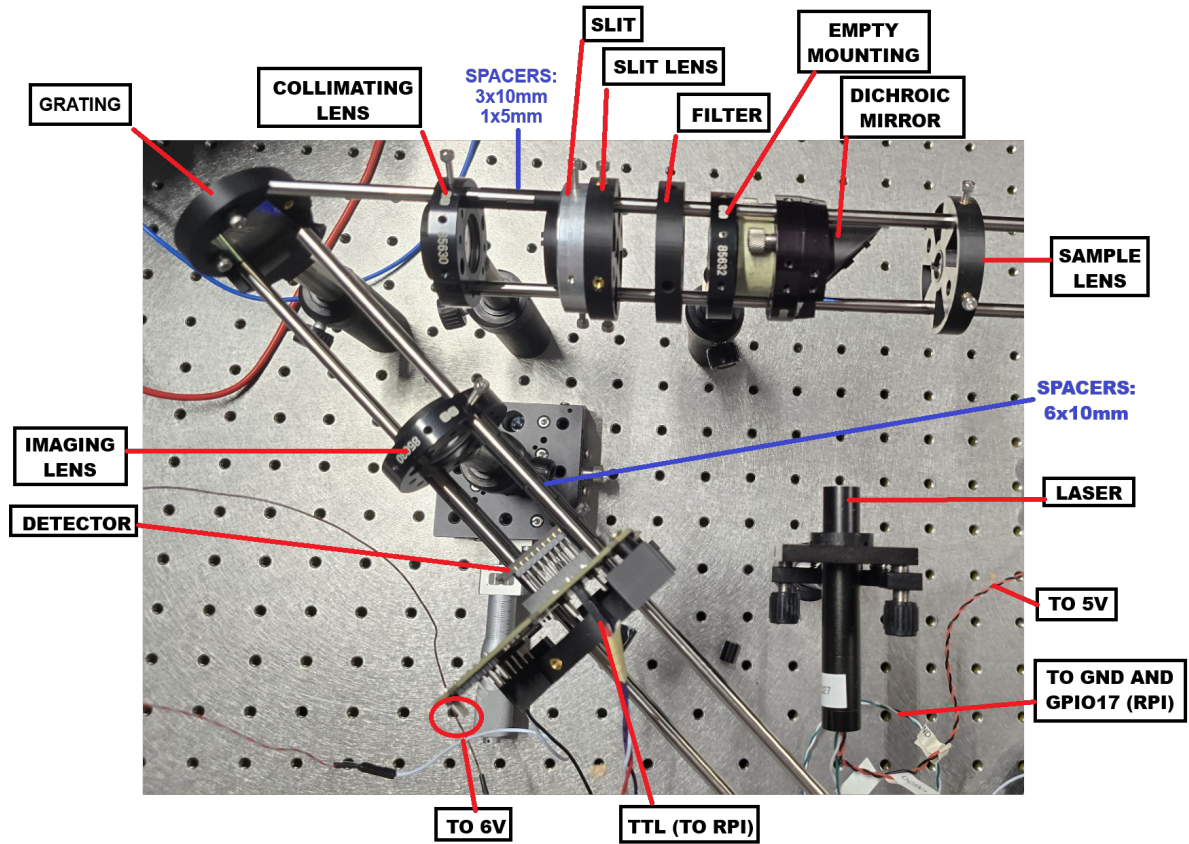


Figure 14: Final optomechanical assembly of the Raman spectrometer prototype. Red indicators identify the main optical elements, while blue annotations indicate the 3D-printed spacers used to reproduce the experimentally adjusted distances.

3. Acquisition, Control and Processing

3.1 Microcontroller selection

For this project, the use of the **STM32F446RCT6** microcontroller in an LQFP64 package was proposed for the final-version design. It should be noted that, during the research and prototyping stage, firmware development was carried out using the STM32F469I Discovery evaluation board.

The final selection of the STM32F446 was made through a cross-technical analysis against the options available in the inventory of the university laboratory, based on the following design criteria:

- **Direct Transition and Ecosystem:** Since it belongs to the same family (STM32F4) and has the same core (ARM Cortex-M4 with floating-point unit - FPU) as the microcontroller used in the prototype, firmware portability is practically immediate. Both share the clock-tree architecture, the peripheral structure (Timers, ADC, DMA), and native support from the STMicroelectronics HAL libraries. Choosing other powerful alternatives from the inventory (such as the PIC32 families from Microchip or TMS320 from Texas Instruments) would have required rewriting the entire hardware abstraction layer, increasing development time.

- **Processing and Timing Capability:** Controlling the TCD1304 CCD sensor requires fast clock signals (2 MHz) and integration times controlled by 32-bit timers to avoid overflows. This made the 8- and 16-bit alternatives available in the inventory less convenient, due to their lower timing, DMA, memory, and processing resources compared with the 32-bit options considered.

Likewise, the NUCLEO-F042K6 development board was discarded. This board has a Cortex-M0 core operating at 48 MHz and severely limited resources (32 KB of Flash and few advanced timers), which offered considerably less margin to simultaneously implement control-signal generation, analog sampling at 500 kHz, DMA transfers, and the serial communication required by the system.

- **Balance between Package and Peripherals:** The inventory included 32-bit alternatives with high capability, but in large packages (e.g., TMS320F28379D in 337-BGA or STM32L4R7VIT6 in 100-LQFP). These packages add unnecessary technical complexity for routing a compact 4-layer PCB aimed at portability. Conversely, other ST alternatives (such as the 20-pin STM32G031F8P6 or 32-pin STM32L081KZT6) did not offer the minimum number of pins required (GPIOs) to interconnect the sensor, Peltier module, UART/I2C communications, and debug signals. The **LQFP64** package of the STM32F446 offers the optimal balance: enough pins for all interfaces without oversizing the board.
- **RAM and Flash Memory:** With 256 KB of Flash memory and 128 KB of SRAM, the STM32F446 has enough capacity to store the timing tables used to generate the SH and ICG signals, the buffers required to acquire the 3694 values of each frame, and the structures required by the communication protocol. Processing of multiple spectra is subsequently delegated to the Raspberry Pi, reducing the memory requirements of the microcontroller.

Once the component was selected, its terminals were assigned. Figure 15 (Table 1) presents the functional diagram of the pins used to interconnect the different modules of the instrument.

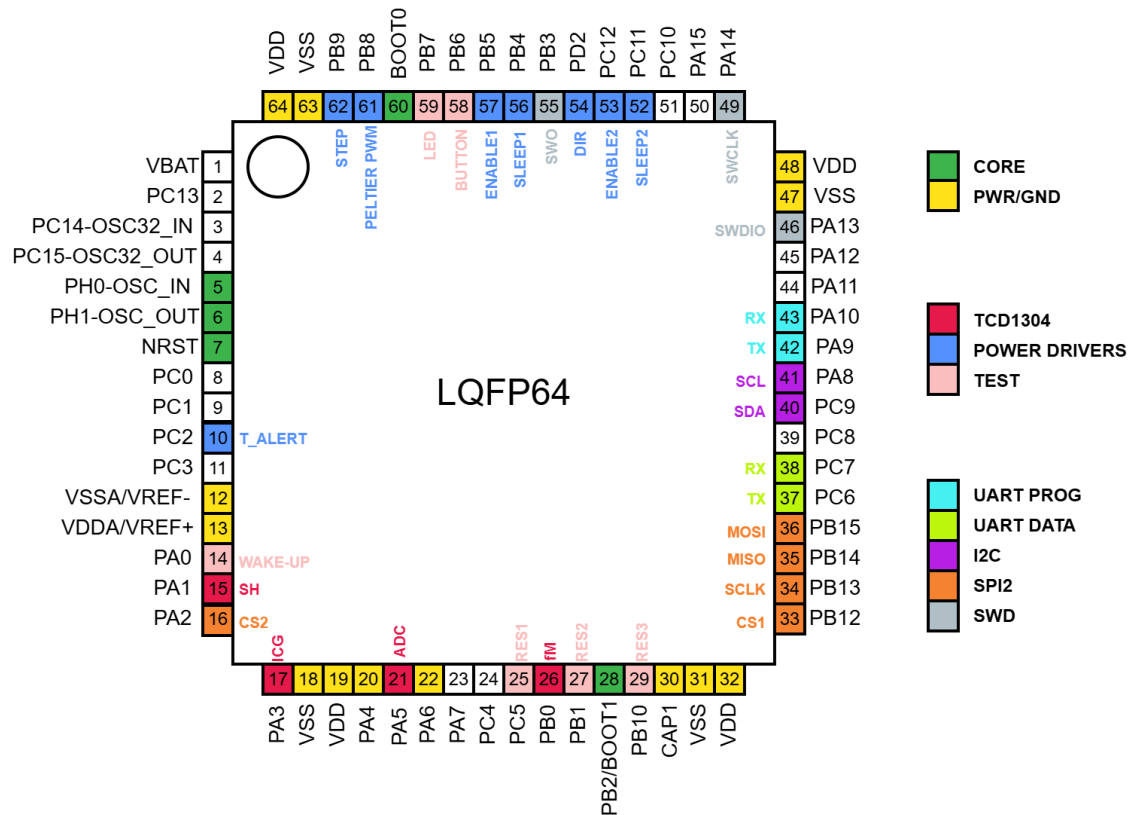


Figure 15: Diagram of the pins used on the STM32F446.

Table 1: Summary of the pins used

Pin	Tag	Function	Type
14	A0	GPIO	WAKE-UP INTERRUPTION
15	A1	TIM5_CH2	SH
16	A2	GPIO	CS2 (SPI)
17	A3	TIM5_CH4	ICG
20	A4	AGND	ADC SHIELD
21	A5	ADC12_IN5	ADC1
22	A6	AGND	ADC SHIELD
41	A8	I2C3_SCL	I2C
42	A9	USART1_TX	UART PROG
<i>Continued on the next page...</i>			

Pin (cont.)	Tag (cont.)	Function (cont.)	Type (cont.)
43	A10	USART1_RX	UART PROG
46	A13	SWDIO	SWD
49	A14	SWCLK	SWD
26	B0	TIM3_CH3	fM
27	B1	GPIO	RESERVED 2
28	B2	BOOT1	BOOT1
55	B3	SWO	SWD
56	B4	GPIO	SLEEP1 (MOTOR)
57	B5	GPIO	ENABLE1 (MOTOR)
58	B6	GPIO	BUTTON (TEST)
59	B7	GPIO	LED (TEST)
61	B8	TIM10_CH1	PELTIER PWM
62	B9	TIM4_CH4	STEP
29	B10	TIM2_CH3	RESERVED 3
33	B12	SPI2_NSS	CS1 (SPI)
34	B13	SPI2_SCK	SCLK (SPI)
35	B14	SPI2_MISO	MISO (SPI)
36	B15	SPI2_MOSI	MOSI (SPI)
10	C2	GPIO	TEMP. ALERT
25	C5	GPIO	RESERVED 1
37	C6	USART6_TX	UART DATA
38	C7	USART6_RX	UART DATA
40	C9	I2C3_SDA	I2C
52	C11	GPIO	SLEEP2 (MOTOR)
53	C12	GPIO	ENABLE2 (MOTOR)
54	D2	GPIO	DIR (MOTOR)
60	BOOT0	BOOT0	BOOT0
7	NRST	NRST	NRST
41	A8	TIM1_CH1 (ALT.)	ADC TRIGGER

To provide the physical design with flexibility during the development and programming stage, the **BOOT0** and **BOOT1** pins (responsible for defining the memory space executed by the core at power-up) were routed to *jumper*-type connectors, enabling manual hardware selection of the different boot modes detailed in Table 2.

Regarding processing speed, the microcontroller has a maximum clock frequency (HCLK) of 180 MHz, which was configured by default for the tests in this project. However, the lower limit of this frequency was analyzed with a view toward future power-

BOOT0	BOOT1	Boot mode (memory selected as boot space)
0	X	Main Flash memory
1	0	System memory
1	1	Embedded SRAM

Table 2: Boot modes (Table 6, ST AN4488)

consumption optimizations. Three hardware restrictions that impose a lower bound on the system frequency were identified:

- The timer responsible for this signal (TIM3) must operate at 2 MHz. To achieve a symmetric duty cycle, the ARR register must have a value of at least 1. Considering that the frequency at which a timer counts is given by:

$$f_{\text{CNT}} = \frac{f_{\text{TIMCLK}}}{\text{PSC} + 1}$$

and that, since a timer counts from 0 to ARR, a complete cycle takes $\text{ARR} + 1$ ticks, therefore:

$$f_{\text{PWM}} = \frac{f_{\text{CNT}}}{\text{ARR} + 1} = \frac{f_{\text{TIMCLK}}}{(\text{PSC} + 1)(\text{ARR} + 1)}$$

Therefore, the minimum TIMCLK frequency required is

$$f_{\text{TIMCLK}} = f_{\text{PWM}}(\text{PSC} + 1)(\text{ARR} + 1) = 2 \text{ MHz} \cdot 1 \cdot 2 = 4 \text{ MHz}$$

- The ADC requires a minimum frequency of 0.6 MHz (Table 74 - STM32F446xC/E datasheet). Using the minimum prescalers, this frequency requires PCLK2 to operate at 1.2 MHz.

3.2 Acquisition and timing strategy

3.2.1 Peripheral configuration

The TCD1304 output signal is digitized using the microcontroller's ADC1, using channel 5 corresponding to pin PA5 (ADC12_IN5). Table 3 summarizes the configuration used.

The converter operates in independent mode, with 12-bit resolution and right-aligned data. Since only one analog channel is used, Scan mode as well as continuous and discontinuous conversion modes were disabled. Instead, each conversion is started by an external event generated by channel 1 of TIM1.

The ADC clock is obtained from PCLK2 through a prescaler of 4. Since PCLK2 operates at 90 MHz, the resulting peripheral frequency is:

$$f_{\text{ADC}} = \frac{90 \text{ MHz}}{4} = 22.5 \text{ MHz}$$

Parameter	Configuration
Mode	Independent mode
Channel	ADC1 Channel 5 (PA5)
Clock Prescaler	PCLK2 divided by 4
ADC Clock	22.5 MHz
Resolution	12 bits
Data Alignment	Right alignment
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Enabled
End of Conversion Selection	End of each conversion
Number of Conversions	1
External Trigger Source	TIM1 Capture Compare 1 event
External Trigger Edge	Rising edge
Sampling Time	28 ADC clock cycles

Table 3: ADC1 configuration.

Continuous DMA requests were enabled to automatically transfer each result from the ADC data register to memory. In this way, once acquisition of a frame begins, the samples can be stored consecutively without CPU intervention.

The result of each 12-bit conversion is stored in 16-bit words. A total of 3694 samples are acquired for each frame, corresponding to the data set subsequently transferred to the processing system.

3.2.2 Timing and sampling

The TCD1304 video signal is sampled at a frequency of 500 kHz, equivalent to one sample every:

$$T_s = \frac{1}{500 \text{ kHz}} = 2 \mu\text{s}$$

Therefore, each ADC conversion must finish within this time window.

For 12-bit resolution, the total time required for a conversion is determined by the sum of the sampling time and the conversion time:

$$T_{\text{ADC}} = \frac{N_{\text{sampling}} + 12}{f_{\text{ADC}}}$$

where N_{sampling} is the number of cycles selected for the sample-and-hold stage.

The STM32F446 allows sampling times of 3, 15, 28, 56, 84, 112, 144, or 480 cycles to be selected. For an ADC frequency of 22.5 MHz, the values compatible with a new sample every 2 μs are 3, 15, and 28 cycles.

28 cycles were selected, since they represent the longest available sampling time that allows the conversion to be completed before the next trigger event:

$$\begin{aligned} T_{\text{ADC}} &= \frac{28 + 12}{22.5 \text{ MHz}} \\ &\approx 1.78 \mu\text{s} \end{aligned}$$

This value leaves an approximate margin of:

$$2 \mu\text{s} - 1.78 \mu\text{s} \approx 0.22 \mu\text{s}$$

before the next trigger.

Selecting a relatively long sampling time also favors proper settling of the voltage on the ADC's internal sample-and-hold capacitor, reducing sensitivity to the analog source impedance.

Considering the 3694 samples acquired per frame, the total time required to complete its readout is:

$$\begin{aligned} T_{\text{readout}} &= 3694 \cdot 2 \mu\text{s} \\ &= 7388 \mu\text{s} \\ &\approx 7.4 \text{ ms} \end{aligned}$$

This readout time is one of the main constraints for generating the SH and ICG signals, as detailed in the timer architecture section.

3.2.3 Synchronization and data transfer

Acquisition is performed through pin PA5 (ADC12_IN5), while TIM1 is used as the periodic trigger source for the ADC.

TIM1 generates events at a frequency of 500 kHz through its channel 1. In turn, it is configured as a slave of TIM5 using the internal ITR0 input. TIM5 generates the ICG signal through its channel 4 and uses the reference of this channel (OC4REF) as the TRGO signal toward TIM1.

This relationship makes it possible to synchronize the beginning of the sampling sequence with the ICG timing, keeping the ADC readout aligned with the CCD data-output window.

The continuous generation of timing signals does not imply that all frames are necessarily stored. In the callback associated with channel 4 of TIM5, the firmware first checks the conditions required to start a new acquisition. In particular:

- the frame-skip counter (`skip_counter`) is checked;
- it is verified through the `adc_busy` flag that the ADC is not performing a previous acquisition;
- it is verified through `can_save` that the previous frame is no longer pending transmission.

When these conditions are met, the complete acquisition is started using:

```
1 HAL_ADC_Start_DMA(&hadc1, (uint32_t*)adc_buffer, CCD_PIXELS);
```

The DMA automatically stores the 3694 consecutive conversions in the `adc_buffer` array. The transfer is performed by incrementing the memory address and using 16-bit words, which are suitable for storing the 12-bit results delivered by the ADC.

Once the transfer is complete, the ADC completion callback is executed. The acquired frame is then available for subsequent transmission to the Raspberry Pi, while the synchronization mechanism prevents the buffer from being reused before the previous data have been sent.

In this way, acquisition timing is mainly handled by hardware peripherals, while the firmware only supervises the beginning of each frame and manages resource availability.

3.3 Timer architecture

Operation of the TCD1304 sensor requires synchronized generation of the f_M , SH, and ICG signals, as well as the periodic signal used to trigger ADC conversions. To minimize processor intervention and guarantee deterministic timing relationships, these signals are generated through an architecture based on the TIM3, TIM5, and TIM1 timers and their internal Master-Slave connections.

Timing starts from the microcontroller clock tree. The system operates with a system frequency of 180 MHz. The APB1 and APB2 buses use prescalers of 4 and 2, respectively:

$$f_{\text{PCLK1}} = \frac{180 \text{ MHz}}{4} = 45 \text{ MHz}$$

$$f_{\text{PCLK2}} = \frac{180 \text{ MHz}}{2} = 90 \text{ MHz}$$

When the prescaler of an APB bus is different from 1, the associated timers receive a clock equivalent to twice the bus frequency. Therefore:

$$f_{\text{TIM,APB1}} = 90 \text{ MHz}$$

$$f_{\text{TIM,APB2}} = 180 \text{ MHz}$$

Thus, TIM3 and TIM5, which belong to APB1, receive a 90 MHz clock, while TIM1, which belongs to APB2, receives a 180 MHz clock.

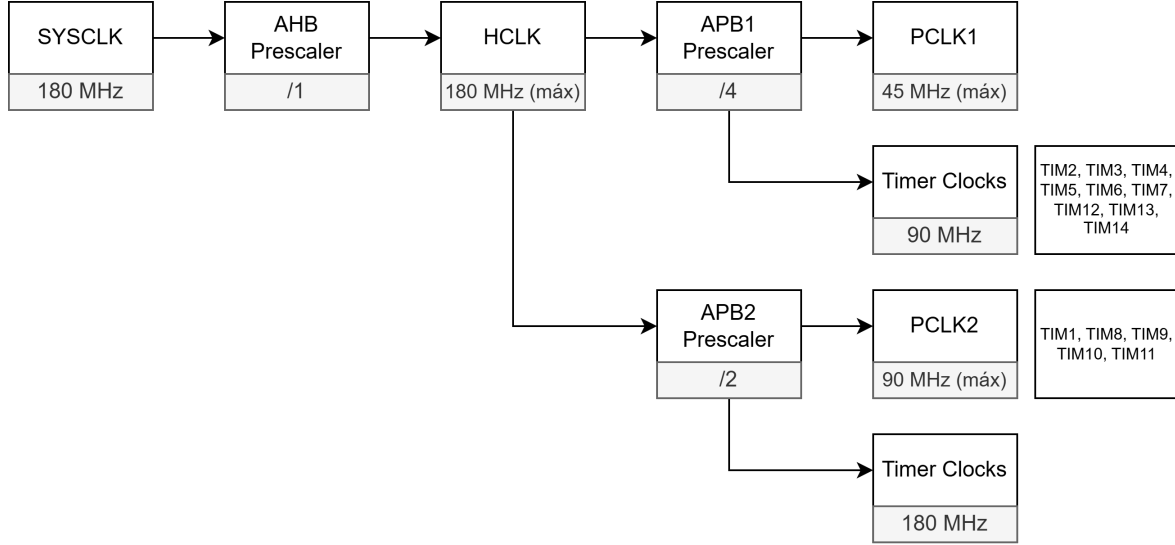


Figure 16: Clock tree used to generate the control signals.

3.3.1 Signal generation and synchronization

The implemented architecture can be summarized as follows:

- **TIM3 – fM generation:** TIM3 is used to generate the main clock signal of the TCD1304, at 2 MHz. The timer operates with its 90 MHz internal clock, without prescaler, and with an ARR value equal to 44:

$$f_{fM} = \frac{90 \text{ MHz}}{(0 + 1)(44 + 1)} = 2 \text{ MHz}$$

Channel 3 operates in PWM1 mode with a compare value of 22, obtaining a duty cycle approximately equal to 50%. The signal is physically available on pin PB0.

In addition to generating the fM output, TIM3 operates as a master and generates a TRGO event with each counter update event. This event is used as the clock input for TIM5, so that the latter increments its counter once for each fM cycle.

- **TIM5 – SH and ICG generation:** TIM5 is a 32-bit timer used to generate the SH and ICG control signals. It was configured as a slave of TIM3 through the internal ITR1 input and in External Clock Mode 1. As a result, its counter advances from the events generated by TIM3 and has a time resolution of:

$$T_{\text{TIM5}} = \frac{1}{2 \text{ MHz}} = 0.5 \mu\text{s}$$

Channel 2 of TIM5 generates SH using Output Compare in Toggle mode. The instants corresponding to each edge are not fixed, but are stored in a table of CCR values calculated by firmware and transferred to the timer through DMA.

Channel 4 generates the ICG signal through PWM and uses its internal OC4REF reference as the TRGO output. This signal is in turn used to synchronize TIM1 with the beginning of the CCD readout window.

The choice of TIM5 is especially convenient because of its 32-bit counter. In this way, integration times of several seconds can be represented directly using a resolution of $0.5 \mu\text{s}$ without the need to dynamically modify the prescaler.

- **TIM1 – ADC trigger generation:** TIM1 is used to generate ADC trigger events at a frequency of 500 kHz. Since it belongs to the APB2 bus, it receives a 180 MHz clock. With a prescaler of 89, a counter frequency of the following is obtained:

$$f_{\text{CNT,TIM1}} = \frac{180 \text{ MHz}}{89 + 1} = 2 \text{ MHz}$$

Using an ARR equal to 3, the channel 1 frequency becomes:

$$f_{\text{ADC trigger}} = \frac{2 \text{ MHz}}{3 + 1} = 500 \text{ kHz}$$

TIM1 is configured as a slave of TIM5 through the ITR0 input and in Trigger mode. In this way, the beginning of the generation of the 500 kHz events is synchronized with the signal coming from the channel used for ICG.

3.3.2 Flushing and integration pulses

As determined in the previous section, reading the 3694 samples from the sensor requires approximately:

$$T_{\text{readout}} \approx 7.4 \text{ ms}$$

This time must be respected regardless of the selected integration time.

A simple implementation would consist of separating all SH pulses using the configured integration time directly. However, for long integration times this would cause the sequence required to complete the CCD readout to be unnecessarily extended. For example, for an integration of several seconds, waiting the full integration time between each pulse required during readout would produce excessive acquisition times.

To avoid this, a sequence of intermediate *flushing* pulses was implemented. During the window required to complete the readout, SH pulses are generated using a fixed period of:

$$T_{\text{flush}} = 100 \mu\text{s}$$

The firmware generates as many flushing intervals as necessary to cover the CCD readout time. Once this sequence is complete, the final interval is extended according to the requested integration time.

In this way, two different constraints are decoupled:

- the time required to shift and completely read the data stored in the CCD;
- the time during which the sensor remains integrating the light corresponding to the next spectrum.

Figure 17 shows an oscilloscope measurement in which the intermediate SH pulses used during flushing and the final interval corresponding to the integration time can be observed.

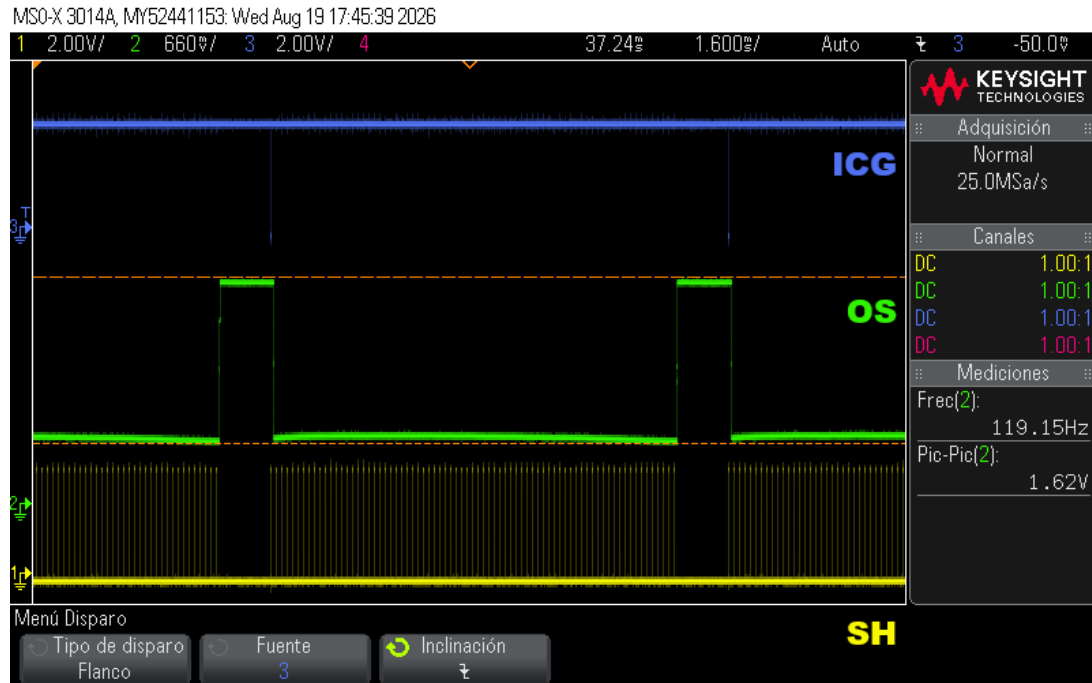


Figure 17: SH signal measured with an oscilloscope. The intermediate flushing pulses, separated by approximately $100 \mu s$, and the final interval determined by the configured integration time (in this case, 1 ms) can be observed.

The exact number of pulses and the positions of each edge are dynamically calculated in firmware and stored in the CCR value table used by TIM5 DMA. The mathematical procedure used to build this sequence is described in the Firmware section.

3.3.3 Configuration summary

Table 4 summarizes the main parameters of the timers used.

	fM	ICG	SH	ADC Trigger
Timer	TIM3	TIM5	TIM5	TIM1
Channel	CH3	CH4	CH2	CH1
Clock / Source	Internal, 90 MHz	TIM3 via ITR1	TIM3 via ITR1	Internal, 180 MHz
Prescaler	0	0	0	89
Counter frequency	90 MHz	2 MHz	2 MHz	2 MHz
ARR	44	Dynamic	Dynamic	3
Mode	PWM1	PWM2	Output Compare Toggle	PWM1
Main parameters	Pulse = 22 Duty cycle $\approx 50\%$	Pulse = 13 TRGO = OC4REF	CCR values loaded by DMA	Pulse = 3 500 kHz
Master/Slave relation	Master TRGO = Update	Slave of TIM3 Master of TIM1	Slave of TIM3	Slave of TIM5 Trigger Mode
Pin	PB0	PA3	PA1	PA8 (not required externally)

Table 4: Summary of the configuration of the timers used to control the TCD1304 and trigger the ADC.

The other specific functions of the advanced TIM1 timer, such as Break, Dead Time, Automatic Output, and the output states during Run and Idle, remain disabled because they are not required for this application.

3.4 Firmware

The microcontroller firmware is responsible for three main functions: generating the timing required by the TCD1304, acquiring each frame through ADC and DMA, and managing communication with the Raspberry Pi through UART.

The implementation was mainly divided into the modules responsible for generating the CCD signals (`tcd_signals`) and serial communication (`tcd_uart`), while the main file initializes the peripherals and coordinates the execution of pending tasks.

3.4.1 SH and ICG timing calculation

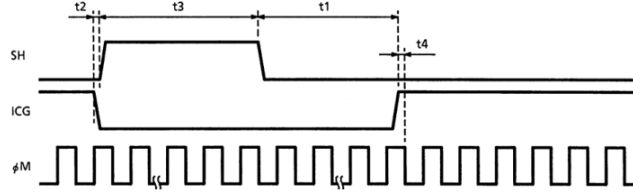
The function

```
1 void calculate_times(uint32_t t_int_us)
```

calculates the times required to generate the SH and ICG signals from the integration time requested by the user.

According to the timing diagram specified in the TCD1304 datasheet, minimum times must be respected between the edges of both signals. These constraints are shown in Figure 18.

Timing Requirements



Characteristic	Symbol	Min	Typ.	Max	Unit
ICG pulse delay	t1	1000	5000	—	ns
Pulse timing of ICG and SH	t2	100	500	1000	ns
SH pulse width	t3	1000	—	—	ns
Pulse timing of ICG and φM	t4	0	20	*	ns

*: Keep the φM pin "H" level.

Figure 18: Timing constraints of the SH and ICG signals specified for the TCD1304.

Since TIM5 increments its counter at a frequency of 2 MHz, each timer tick represents:

$$T_{\text{tick}} = 0.5 \mu\text{s}$$

The implemented sequence is divided into intervals S_0 to S_6 , according to the diagram shown in Figure 19.

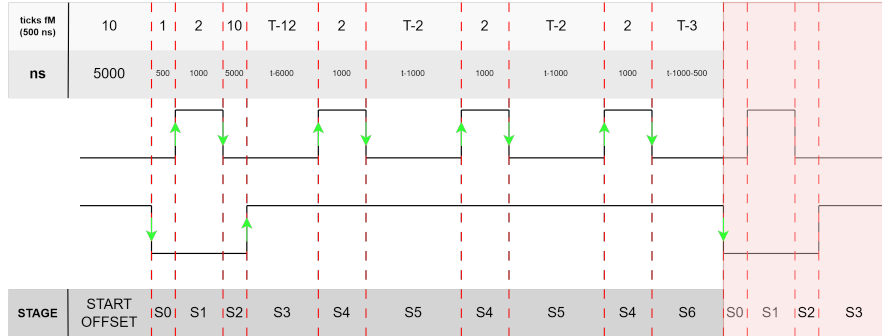


Figure 19: Timing division used by the firmware to generate SH and ICG.

The first three intervals are fixed at compile time:

$$S_0 = 1 \text{ tick} = 0.5 \mu\text{s}$$

$$S_1 = 2 \text{ ticks} = 1 \mu\text{s}$$

$$S_2 = 10 \text{ ticks} = 5 \mu\text{s}$$

As explained previously, during the readout stage a fixed flushing period of:

$$T_{\text{flush}} = 100 \mu\text{s}$$

is used. Therefore, the intervals used during these pulses are calculated as:

$$\begin{aligned} S_3 &= T_{\text{flush}} - S_1 - S_2 \\ S_4 &= S_1 \\ S_5 &= T_{\text{flush}} - S_4 \end{aligned}$$

With the selected values:

$$\begin{aligned} S_3 &= 94 \text{ } \mu\text{s} \\ S_4 &= 1 \text{ } \mu\text{s} \\ S_5 &= 99 \text{ } \mu\text{s} \end{aligned}$$

The final interval of the sequence, on the other hand, depends on the requested integration time:

$$S_6 = t_{\text{int}} - S_0 - S_1$$

The number of intermediate pulses is determined from the minimum time required to complete the sensor readout. For the 3694 samples acquired at 500 kHz:

$$\begin{aligned} T_{\text{readout}} &= \frac{3694}{500 \text{ kHz}} \\ &= 7.388 \text{ ms} \approx 7.4 \text{ ms} \end{aligned}$$

The first falling edge of SH occurs after:

$$T_{\text{first fall}} = S_0 + S_1$$

Therefore, the number of flushing intervals required to cover the remainder of the readout time is calculated by rounding upward:

$$N_{\text{flush}} = \left\lceil \frac{T_{\text{readout}} - (S_0 + S_1)}{T_{\text{flush}}} \right\rceil$$

and the number of intermediate repetitions used by the firmware is:

$$n = N_{\text{flush}} - 1$$

In this way, the number of pulses required during the readout stage depends on the CCD readout time and not on the integration time. This prevents long integrations from unnecessarily increasing the duration of the flushing sequence.

3.4.2 Construction of the SH sequence

Once the times have been calculated, the function

```
1 void build_SH_table(void)
```

constructs the sequence of SH edges.

Initially, a table of relative intervals is generated:

$$[S_0, S_1, S_2 + S_3, S_4, S_5, \dots, S_4]$$

where the S_4 – S_5 pair is repeated n times.

These intervals are then converted into absolute counter positions by a cumulative sum. The result is stored in the `sh_ccr` array:

$$CCR_i = \sum_{k=0}^i \Delta t_k$$

The table is transferred to channel 2 of TIM5 through DMA. Each match between the counter and the CCR value produces a channel state change due to its configuration in Output Compare Toggle mode.

This mechanism makes it possible to generate a sequence with a variable number of edges without the need for software to handle each SH transition.

When the timers are started, TIM5's ARR is also dynamically updated:

```
1 __HAL_TIM_SET_AUTORELOAD(
2     &htim5,
3     sh_ccr[real_SH_EDGES - 1] + TS6_tics
4 );
```

In this way, the total cycle duration incorporates both the flushing sequence and the final interval corresponding to the integration time.

3.4.3 Main program

After initializing the peripherals configured through HAL, the program initially sets an integration time of 100 μ s, calculates the timing, builds the SH table, and starts TIM1, TIM3, and TIM5.

Reception of commands through USART6 is then enabled using interrupts:

```
1 HAL_UART_Receive_IT(
2     &huart6,
3     (uint8_t*)rx_cmd_buffer,
4     SIZE_RX_BUFFER_CMD_BYTES
5 );
```

The main execution uses a `while(1)` loop based on status flags and is mainly limited to supervising two events:

- reception of a new valid instruction, indicated by `process_instruction_flag`;
- existence of a frame ready to be transmitted, indicated by `send_now`.

When an instruction is received, any ADC acquisition in progress is stopped and the program waits for any ongoing UART transmission to finish before modifying the configuration. The corresponding command is then executed, the state variables are reset, and the timers are restarted.

The commands implemented in the final version are:

- microcontroller reset;

- integration-time configuration;
- configuration of the number of frames to skip;
- test LED toggling.

Unlike earlier versions of the firmware, accumulation and averaging of multiple spectra are not performed on the microcontroller. Each frame is transmitted individually to the Raspberry Pi, where subsequent processing is performed.

3.4.4 Acquisition using callbacks

The start of an acquisition is managed from:

```
1 HAL_TIM_PWM_PulseFinishedCallback(...)
```

associated with channel 4 of TIM5.

Before starting the ADC, three conditions are checked. First, the `skip_counter` counter allows a configurable number of frames to be skipped. This functionality can be used to reduce the effective rate at which spectra are stored and transmitted without modifying the continuous generation of the CCD control signals. Second, the `adc_busy` flag is checked, preventing a new acquisition from being started while the ADC is still completing the previous frame.

Finally, the `can_save` flag indicates whether the buffer containing the last frame can already be reused. If the previous frame is still being transmitted through UART, an acquisition that could overwrite its data is prevented from starting.

When the conditions are met, the following is executed:

```
1 adc_busy = 1;
2 HAL_ADC_Start_DMA(
3     &hadc1,
4     (uint32_t*)adc_buffer,
5     CCD_PIXELS
6 );
```

From that point on, the 3694 samples are automatically stored in `adc_buffer`.

When the transfer is complete, the following is executed:

```
1 HAL_ADC_ConvCpltCallback(...)
```

The acquired content is copied to the `frame` array, the `adc_busy` flag is released, the possibility of overwriting the frame is temporarily blocked through `can_save = 0`, and `send_now` is activated to indicate to the main program that data are available for transmission.

3.4.5 UART communication protocol

Communication between the STM32 and the Raspberry Pi is carried out through USART6 at a rate of 921600 baud.

Commands sent from the Raspberry Pi use 10-byte packets organized in 16-bit words:

$$[\text{HEADER}][\text{COMMAND}][\text{PAYLOAD}_L][\text{PAYLOAD}_H][\text{CHECKSUM}]$$

The payload occupies 32 bits and allows, for example, the integration time expressed in microseconds to be transmitted directly.

Command integrity is verified by means of an XOR checksum applied to all received words. Since the transmitted checksum is part of the same operation, a valid packet satisfies:

$$W_0 \oplus W_1 \oplus W_2 \oplus W_3 \oplus W_4 = 0$$

If the header and checksum are valid, the command and payload are extracted and `process_instruction_flag` is activated, leaving the actual processing outside the interrupt.

For commands that are correctly applied, an ACK confirmation frame is transmitted before restarting the timing signals.

3.4.6 Spectrum transmission

Each acquired frame is transmitted using DMA in order to avoid keeping the CPU busy during serial communication.

The data frame is composed of:

[HEADER][DATA_SENDING][PIXEL₀] ... [PIXEL₃₆₉₃][CHECKSUM][END_BUFFER]

Each element occupies 16 bits, so the complete packet contains:

$$3694 + 4 = 3698$$

16-bit words, equivalent to:

$$7396 \text{ bytes}$$

USART6 uses an 8N1 configuration, so each byte requires 10 bits on the line. The approximate time required to transmit a complete frame at 921600 baud is therefore:

$$T_{\text{UART}} = \frac{7396 \cdot 10}{921600} \\ \approx 80.3 \text{ ms}$$

This time is considerably longer than the approximately 7.4 ms required to digitize one frame. For this reason, acquisition and transmission cannot simultaneously use the same buffer without some protection mechanism.

The `can_save` flag acts as a semaphore between both operations. Once acquisition is complete, it is set to zero, preventing the contents of `frame` from being overwritten. Transmission is started using:

```
1 HAL_UART_Transmit_DMA (
2     &huart6,
3     (uint8_t*)tx_packet_buffer,
4     bytes_to_send
5 );
```

When the DMA finishes, the following is executed:

```
1 HAL_UART_TxCpltCallback(...)
```

where `uart_busy` is released and the following is set again:

```
1 can_save = 1;
```

enabling the capture of a new frame.

This strategy prioritizes data integrity over uninterrupted acquisition: if transmission of the previous spectrum has not yet finished, the next available frame is simply not stored. The sensor continues operating normally, and a new acquisition is started when the buffer becomes available again.

4. Hardware

The physical design was implemented on two 4-layer PCBs, with the internal layers mainly allocated to power and ground planes in order to reduce parasitic inductances and improve electromagnetic compatibility (EMC).

In order to reduce possible noise coupling from higher-power devices onto the sensitive acquisition stages, the system was divided into two boards. This separation makes it possible to physically isolate the electronics associated with the detector and signal processing from the complementary power blocks. The following subsections describe the design of each board.

4.1 Power-supply conditioning

4.1.1 Bypass capacitors

In areas close to power supplies or integrated circuits, networks of parallel capacitors were incorporated. Although, for a digital switching event, a single $0.1\ \mu\text{F}$ capacitor could store much more charge than required to compensate for current peaks, real components exhibit ESR (Equivalent Series Resistance) and ESL (Equivalent Series Inductance). These parameters, which depend on material and package, modify frequency behavior: while an ideal capacitor would reduce its impedance linearly, in a real one ESL begins to dominate beyond a certain point, causing the impedance to rise again and making it necessary to compensate for this with other components.

Ceramic capacitors are preferred because of their low ESR and ESL, while tantalum capacitors offer a high capacitance-to-volume ratio with moderate ESR/ESL, ideal for low-frequency variations. Package size is decisive: the larger the package, the greater the ESL. For example, a $1\ \mu\text{F}$ tantalum capacitor has a lower initial impedance than a 0603 $0.1\ \mu\text{F}$ ceramic capacitor because of its higher capacitance, but its higher ESR/ESL causes its response to degrade above 100 kHz (Figure 20)¹. At 10 MHz, the ceramic capacitor may exhibit an impedance ten times lower than the tantalum capacitor. The design solution is to place these three types in parallel: by combining them, the total impedance of the network is determined by the component that presents the lowest value at each specific frequency, achieving a constant low-impedance path over a very wide frequency spectrum.

4.1.2 Ferrite bead

Definition and operating principle. The simplest form of a ferrite bead consists of a conducting wire inserted through a hollow ceramic core. The electromagnetic properties of this material influence the current flowing through the conductor, depending on the type of ferrite and the winding construction. In the context of power-supply filtering, resonance is to be minimized, so low-Q (non-resonant) beads are used. As shown in the first-order model in Figure 21, the predominant response is inductive at low frequencies (Z increases with f). However, at a transition point (typically between 30 MHz and 500 MHz), parasitic capacitance begins to dominate and the impedance decreases. The parasitic resistance ($R_{par} \approx 100\ \Omega$) is responsible for reducing resonance, stabilizing the impedance instead of generating high-Q peaks.²

¹All About Circuits - Clean Power for Every IC

²Understanding Ferrite Beads

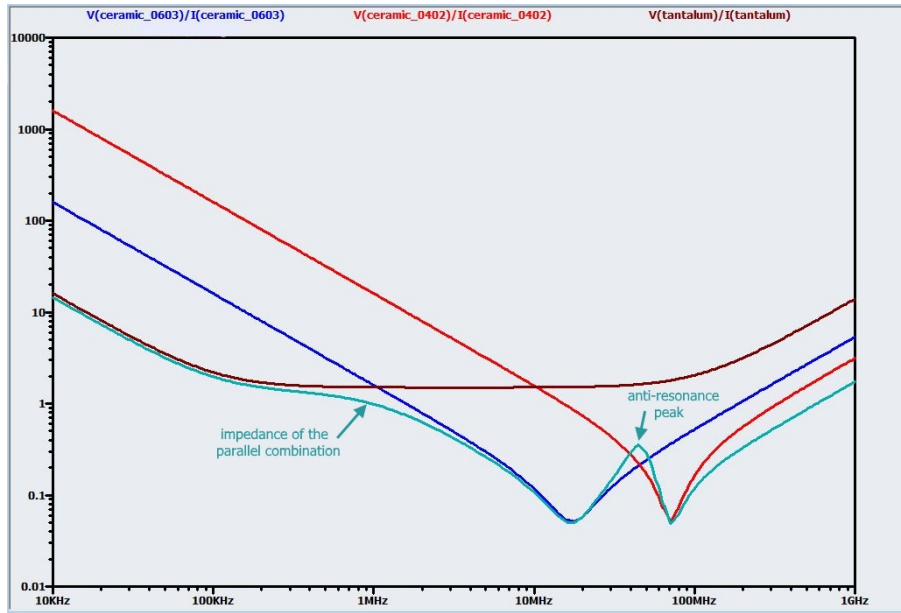


Figure 20: Frequency response for capacitors of different materials and packages

Energy storage and dissipation. Unlike ideal inductors and capacitors, which only store energy in magnetic or electric fields, the ferrite bead is intentionally designed to be resistive at high frequencies. Figure 22 shows the characteristic curve of a standard SMD bead:

- Low frequencies: It starts at R_{series} (R_{dc}) and increases linearly (reactive/inductive behavior, X_L)
- High frequencies (100 MHz to 1 GHz): The resistive impedance (R) predominates over the reactive impedance (X_L)

This “lossy” characteristic makes it possible to dissipate noise as heat instead of storing it. While a conventional inductor can interact with parasitic PCB capacitances to create resonant LC circuits and generate ringing, the ferrite damps these oscillations, providing low DC resistance and high dissipation in the target noise band.

Selection criteria. To ensure the effectiveness of the bead, three critical factors must be considered³:

- Voltage drop (R_{dc}): Since it is a series component, DC current generates a drop $V = I \times R_{dc}$. In low-voltage systems, this can be critical. For example, on a 1.1 V rail, a bead with $R_{dc} = 0.7 \Omega$ under a 400 mA current peak would reduce the voltage to 0.82 V, causing intermittent failures that are difficult to diagnose
- Rated Current and Saturation: The ferrite material saturates with DC current, which reduces its maximum impedance and shifts the curve toward higher frequencies. It is recommended that the rated current be at least 50% or 100% higher than the expected maximum current. Temperature derating must also be considered (e.g., abrupt reductions above 85°C)

³Choosing and Using Ferrite Beads

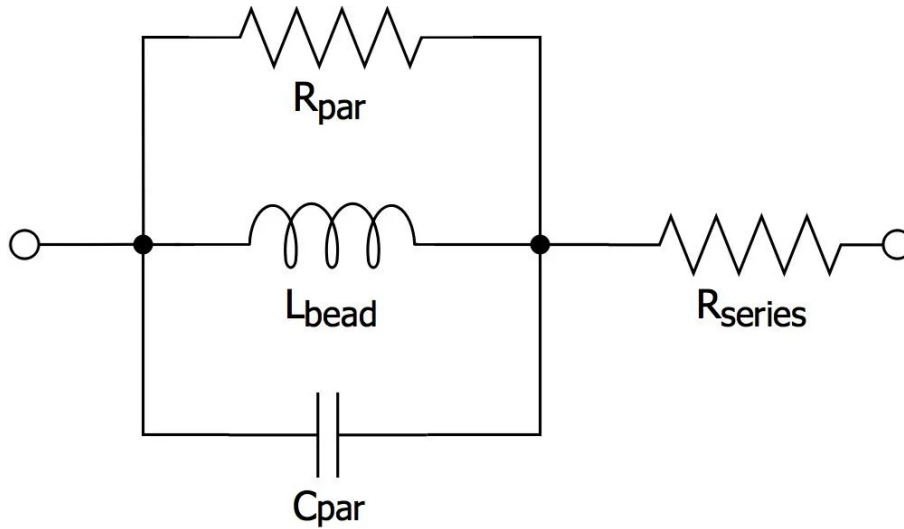


Figure 21: First-order model of the ferrite bead

- Resonance with bypass capacitors: If large capacitors ($> 10 \mu\text{F}$) are used, the system resonance frequency may fall within the reactive band of the bead, potentially generating oscillations if there is not enough parasitic resistance to damp them.

PCB implementation. Since an inductor opposes abrupt changes in current, placing a bead directly before the power pin without a reservoir capacitor (Figure 23) would block the transient current required by the IC.

The correct configuration requires placing the bypass capacitor downstream of the ferrite bead (Figure 24). In this way, the IC obtains the transient current directly from the capacitor, while the bead isolates noise between stages, allowing, for example, analog and digital sections to be powered from the same regulator with adequate isolation.

Selection. Two critical parameters were analyzed for selecting the ferrite beads: the operating frequency of the system and the current consumption of the components to be filtered.

An isolation ferrite bead was placed between the V_{DD} (Digital) and V_{DDA} (Analog) nets. According to the microcontroller power architecture, the V_{DDA} pin supplies power to the ADC, DAC, reset blocks, internal oscillators (RCs), and the PLL circuit⁴. The estimated maximum consumption for this domain is broken down below:

- ADC: 1.8 mA (Maximum consumption in conversion mode⁵)
- DAC: 0 mA (Peripheral disabled in firmware)
- Reset Blocks: ~ 0.1 mA (Voltage supervision circuits)
- RC Oscillators: 0 mA (An external HSE crystal is used, keeping the internal oscillators off)
- PLL: 0.85 mA [⁶]

⁴STM32F446 - 3.15 Power supply schemes" (p. 22)

⁵STM32F446 - Table 74: ADC VDDA DC current consumption in conversion mode (max)

⁶STM32F446 - Table 43: PLL Power consumption on VDDA (max)

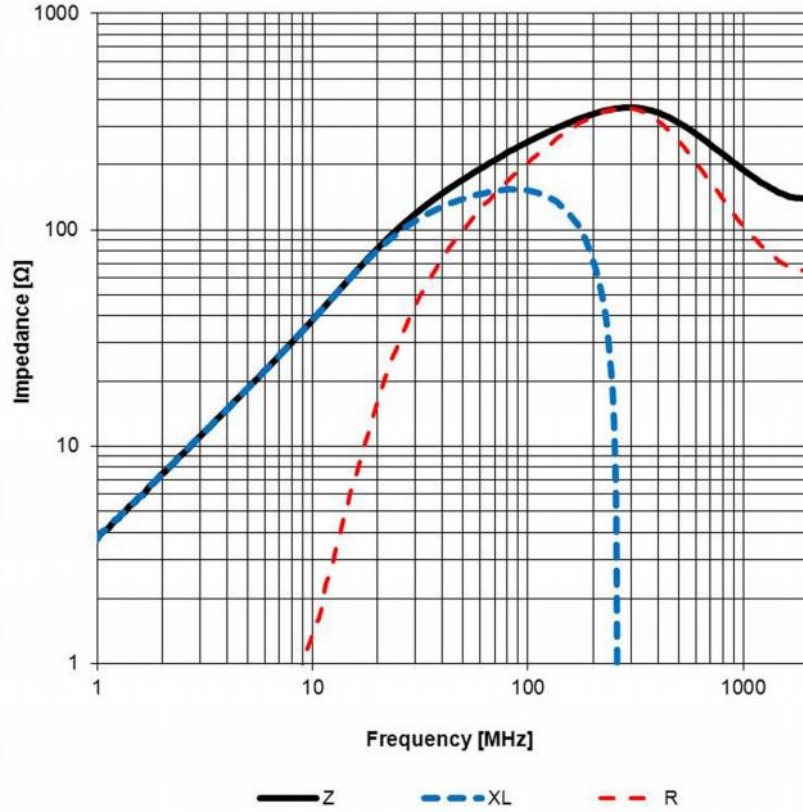


Figure 22: Ferrite-bead impedance as a function of frequency

The total consumption of the analog domain is approximately 2.75 mA. From a strictly magnetic point of view, a rated current of 5 or 10 mA would be sufficient, although the impedance curves still need to be analyzed.

The MMZ1608B601CTAH0 ferrite, available in the laboratory, was selected. It has a maximum DC resistance (R_{DC}) of $0.40 \, \Omega$ and a rated current of 500 mA. With the calculated consumption, the maximum voltage drop is $\sim 2.75 \, \text{mA} \cdot 0.4 \, \Omega = 1.1 \, \text{mV}$.

Analysis of Figure 25 [7] places the component impedance at $600 \, \Omega$ at 90 MHz (APB1), reaching its maximum peak around 100 – 200 MHz (useful for the 180 MHz of APB2). In addition, the reactance decreases considerably in this frequency range, especially above 100 MHz, allowing switching noise to be dissipated without generating unwanted resonances on the analog lines.

For the TCD1304DG sensor, the supply of the analog output amplifier ($V_{TCD A}$) was isolated. Since the datasheet does not directly specify the current of this pin, an estimate was made based on the maximum allowed power dissipation of 75 mW. Under a typical supply of 4 V, a maximum current of $I_{MAX} = \frac{75 \, \text{mW}}{4 \, \text{V}} = 18.75 \, \text{mA}$ is estimated.

Again, using the MMZ1608B601CTAH0 ferrite, the resulting voltage drop is approximately 7.5 mV. This attenuation is negligible for sensor operation, allowing the filter to act as a high-impedance barrier against high-frequency noise coming from clock edges and the digital environment of the board, preserving the integrity of the OS video signal.

⁷MMZ1608 type - TDK

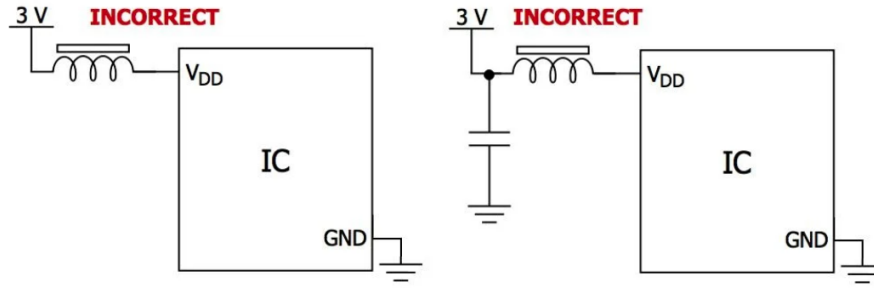


Figure 23: Incorrect ways of connecting the ferrite bead

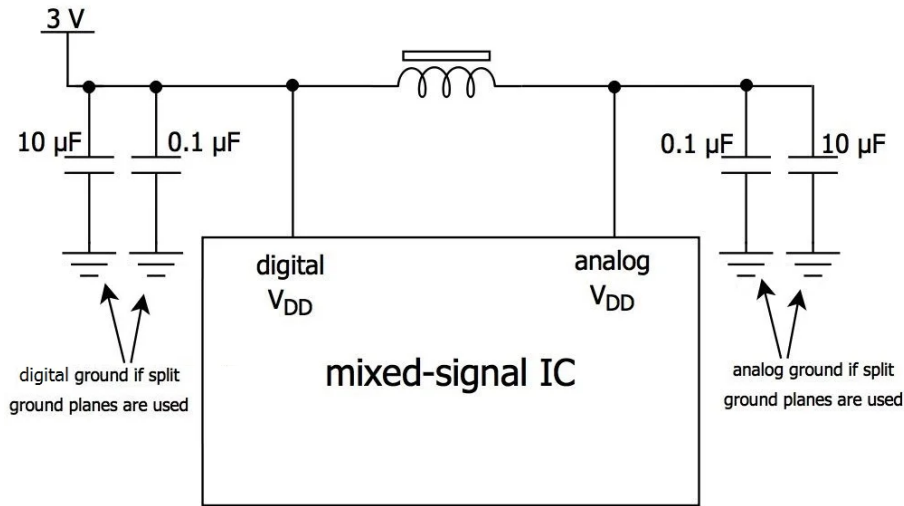


Figure 24: Correct way of connecting the ferrite bead

4.2 PCB A

Board A constitutes the acquisition core of the system and integrates the microcontroller, the CCD sensor, its analog conditioning, the associated power supplies, and the communication and debugging interfaces.

Figure 26 shows the front and rear views of the board design.

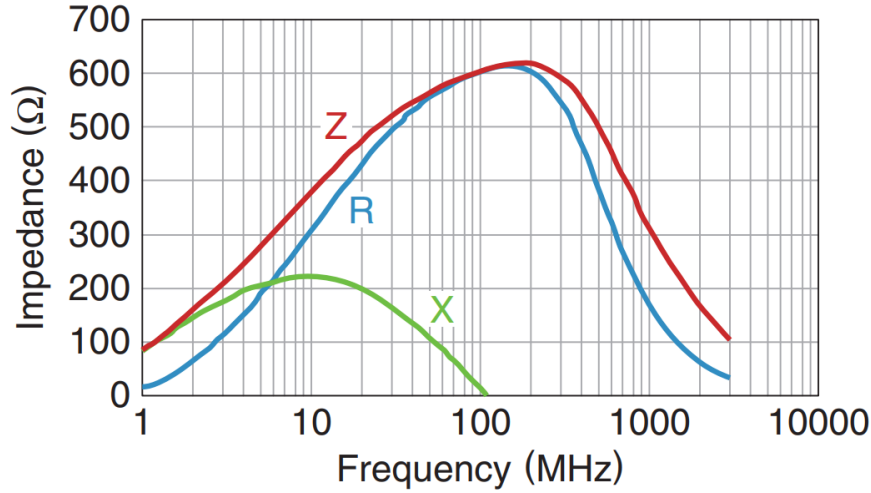


Figure 25: Impedance of MMZ1608B601CTAH0

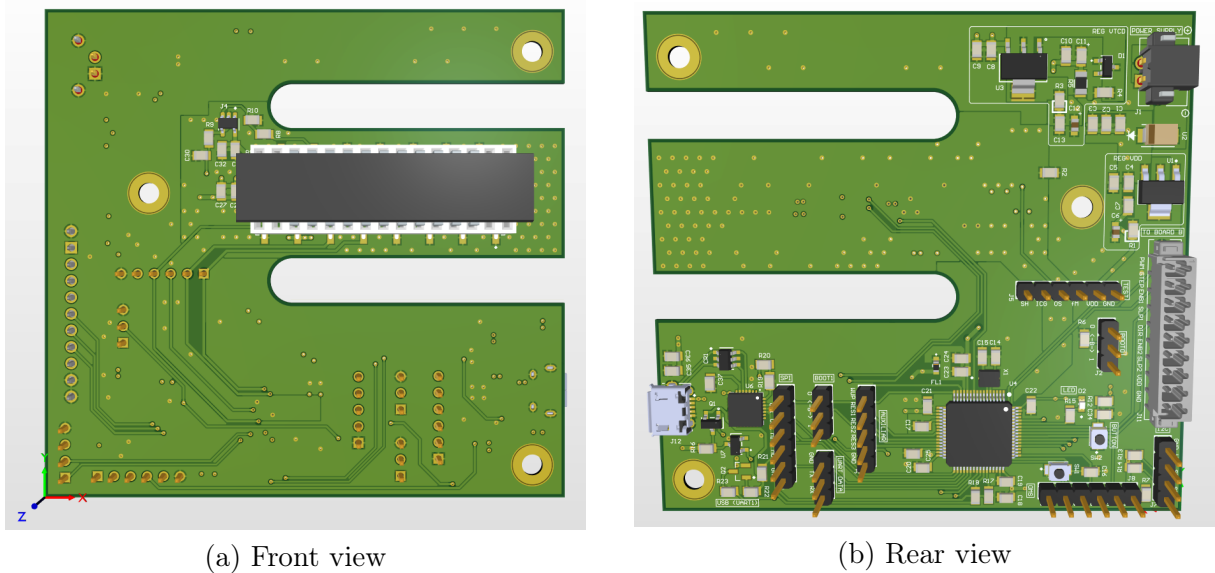
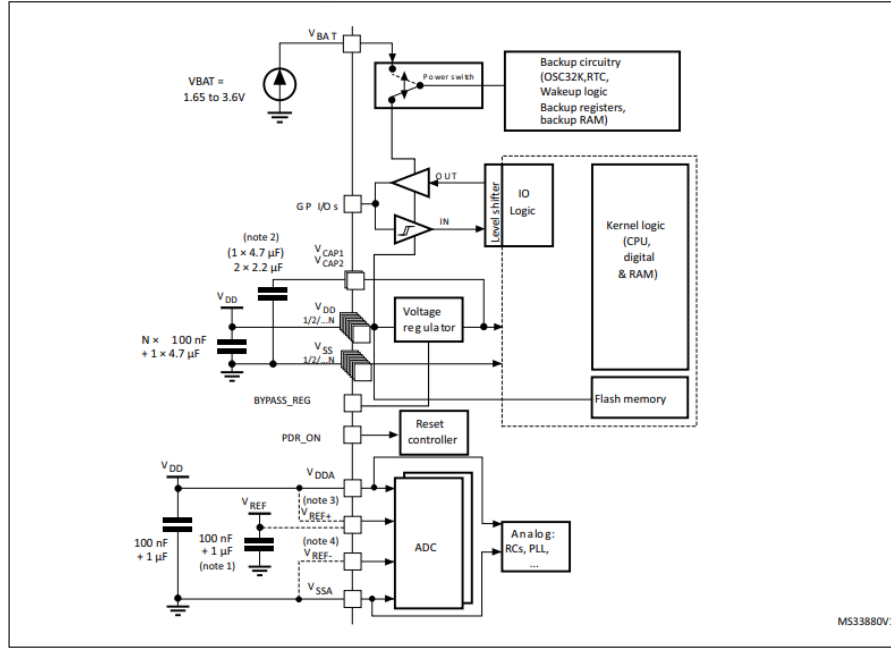


Figure 26: 3D view of PCB A.

4.2.1 Microcontroller

For circuit conditioning, Figure 27, extracted from application note AN4488, was used as a model: a 100 nF capacitor was placed at each VDD input, one of them together with another 4.7 μ F capacitor, one 4.7 μ F capacitor at VCAP1 (whose ESR must be less than 1 Ω), and a 100 nF + 4.7 μ F pair at the analog AVDD input. In addition, following the indications of the NUCLEO-XXXXRX MB1136 datasheet (p. 3), a ferrite core (BEAD) was placed between VDD and AVDD to filter the analog supply and a 100 nF capacitor was added for the reset button. BOOT0 and BOOT1 were placed on pins that allow jumpers to be used to select the state of each one.

Regarding the crystal, a CG04874-8M with 8 pF load capacitance was used and two 10 pF capacitors were placed, based on the following equation (AN2867_rev24, p.12):



1. Optional. If a separate, external reference voltage is connected on V_{REF+} , the two capacitors (100 nF and 1 μ F) must be connected.
2. V_{CAP2} is not available on all packages. In that case, a single 4.7 μ F (ESR < 1 Ω) is connected to V_{CAP1} .
3. V_{REF+} is either connected to V_{REF+} or to V_{DDA} (depending on package).
4. V_{REF-} is either connected to V_{REF-} or to V_{SSA} (depending on package).
5. N is the number of V_{DD} and V_{SS} inputs.
6. Refer to section "Voltage regulator" in datasheet (Table 1) to connect BYPASS_REG and PDR_ON pins.

Figure 27: Power supply scheme

$$C_L = \frac{C_{L1}C_{L2}}{C_{L1} + C_{L2}} + C_S$$

$$8 \text{ pF} = \frac{C}{2} + 3 \text{ pF}$$

$$C = 10 \text{ pF}$$

where C_L is the load capacitance required by the crystal and $C_{L1} = C_{L2} = C$. Since a low-impedance routing strategy was chosen, placing the oscillator crystal and its load capacitors at the minimum possible distance from the microcontroller OSC_IN and OSC_OUT pins, it is assumed that the parasitic capacitance of the PCB is minimized, and an estimated value of 3 pF is proposed.

4.2.2 Conditioning

To optimize the signal-to-noise ratio (SNR) in the acquisition stage, the hardware design incorporated some guidelines stipulated in STMicroelectronics application note AN2834:

Recommendation	Implementation
Use linear regulators and connect filter capacitors directly to their output to minimize power-supply noise	NCV1117 family linear regulators are used to power the system
Connect capacitors with good high-frequency characteristics between the power and ground lines (ceramic capacitors from 1 pF to 0.1 μ F) as close as possible to the microcontroller pins. Additionally, use larger capacitors (10 μ F to 100 μ F) to filter low-frequency noise	All voltage inputs include capacitive tanks consisting of parallel triplets of 100 pF, 100 nF, and 4.7 μ F
Use ferrite inductances in series with the power supply to attenuate high-frequency noise	Ferrite beads were added to the analog power-supply lines
The analog supply pin must be connected to two external decoupling capacitors (10 nF ceramic + 1 μ F tantalum or ceramic)	At the output of the ferrite corresponding to the VDDA pin, a 4.7 μ F and 100 nF capacitor pair was implemented
Ensure that the output resistance of the circuit being measured is below the theoretical limit to guarantee 12-bit accuracy: $R_{\text{AIN,max}} = \frac{T_s}{f_{\text{ADC}} C_{\text{ADC}} \ln(2^{N+1})} - R_{\text{ADC}}$	For a sampling time of 28 cycles, $f_{\text{ADC}} = 22.5 \text{ MHz}$, $C_{\text{ADC}} = 7 \text{ pF}$, $R_{\text{ADC}} = 6 \text{ k}\Omega$ (maximum datasheet values), the limit is $13.73 \text{ k}\Omega$. Since the sensor output includes an operational amplifier and a 100Ω series resistor, the actual impedance seen by the ADC is below this limit (even assuming a conservative output-resistance value of 500Ω)
Add an external RC low-pass filter to limit high-frequency noise and avoid aliasing, ensuring that the analog signal frequency satisfies $F_{\text{AIN}} \leq \frac{1}{10 R_{\text{AIN}} (C_{\text{AIN}} + C_{\text{P}})}$	Assuming a conservative trace parasitic capacitance $C_{\text{P}} \approx 10 \text{ pF}$ and a conservative output impedance of 600Ω (including the series resistance), the maximum limit without a filter is 16.7 MHz . If an external 100 pF capacitor were incorporated, the maximum source frequency would be reduced to 1.51 MHz . This value allows the dynamic signal to be sampled while acting as an effective low-pass filter for frequencies in the tens of MHz. It also absorbs the ADC charging transient (whose internal capacitor is 14 times smaller)
Place ground traces alongside sensitive analog signals to provide shielding on the PCB, preventing interference and crosstalk	A guard trace (AGND) was routed on each side of the OS signal coming from the sensor.
<i>Continued on the next page...</i>	

Recommendation (continued)	Implementation (continued)
Separate analog circuits from digital circuits. It is highly recommended to use multilayer PCBs, assigning dedicated layers for power and ground planes, which reduces trace inductance and improves electromagnetic shielding	A 4-layer PCB was designed, allocating one internal layer exclusively to the power plane and another to the general ground plane
Use physically separate planes for analog and digital grounds. The analog ground must be located strictly below the analog circuits	The physical ground plane of the PCB was strategically divided into three isolated polygons: digital ground, analog ground, and power ground
Power the analog and digital sections from independent power sources when there is a significant amount of circuitry external to the microcontroller	Analogously to the ground division, the internal power plane was segmented into general power, digital supply, and analog supply areas.
Connect capacitors and resistors using extremely short traces. Use surface-mount components (SMD) for decoupling, use wide traces for power lines to avoid voltage drops, and ensure that quartz crystals are surrounded by a ground plane and located as close as possible to the microcontroller	Only SMD passive components were used to minimize package parasitic inductances. Likewise, the quartz crystal was placed at the minimum possible distance from the microcontroller oscillator pins, ensuring a continuous and uninterrupted ground plane on the lower layer for proper isolation

4.2.3 Power-supply module

Board A has an input for connecting the VCC supply. The logic supply is regulated using two NCV1117 LDOs, obtaining 3.3 V for the microcontroller and 4.0 V for the TCD1304 CCD. These regulators have a maximum dropout voltage of 1.2 V and a maximum input voltage of 20 V. In this way, 6 V is established as a suitable value for VCC.

At the input, a parallel capacitive network was placed, consisting of a $4.7 \mu F$ capacitor, a $100 nF$ capacitor, and a $100 pF$ capacitor, to filter low-, medium-, and high-frequency noise coming from the line.

A $10 \mu F$ capacitor was placed at the input and output of each regulator. For the input, the datasheet allows the use of ceramic or tantalum capacitors⁸. For the output, on the other hand, a minimum of $4.7 \mu F$ is required, although larger values offer better stability and reduce noise, together with an equivalent series resistance (ESR) between $33 m\Omega$ and 2.2Ω . For this reason, a $10 \mu F$ capacitor was selected.⁹

⁸Input bypass capacitor C_{in} may be required for regulator stability if the device is located more than a few inches from the power source. This capacitor will reduce the circuit's sensitivity when powered from a complex source impedance and significantly enhance the output transient response. The input bypass capacitor should be mounted with the shortest possible track length directly across the regulator's input and ground terminals. A 10 F ceramic or tantalum capacitor should be adequate for most applications (NCP1117/D, p.9)

⁹Frequency compensation for the regulator is provided by capacitor C_{out} and its use is mandatory to ensure output stability. A minimum capacitance value of 4.7 F with an equivalent series resistance (ESR)

The connections were made according to Figures 23 and 24 of the datasheet (see Figure 28). Considering a maximum current $I_{adj} = 120 \mu A$ and $V_{ref}(typ) = 1.25 V$ ¹⁰, for the 4 V regulator the resistance values $R_3 = 220 \Omega$ and $R_4 = 100 \Omega$ were proposed (the datasheet required it to be kept between 100 and 200 Ω)¹¹, since

$$\begin{aligned} V_{out} &= V_{ref} \left(1 + \frac{R_3}{R_4} \right) + I_{adj} R_3 \\ &= 4.0264 V \end{aligned}$$

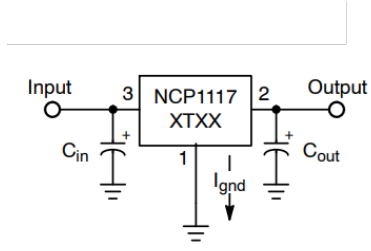


Figure 23. Fixed Output Regulator

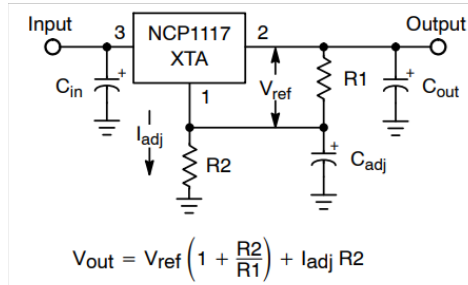


Figure 24. Adjustable Output Regulator

Figure 28: Connections from the NCP1117 datasheet

In addition, considering the ripple produced by the line frequency of the Swiss power grid (50 Hz, 100 Hz when rectified), a 22 μF capacitor (C_6) was placed, satisfying the condition¹²:

$$C_{adj} > \frac{1}{2\pi \cdot 100 \text{ Hz} \cdot R_1} = 15.92 \mu F$$

This capacitor could be multilayer ceramic in order to have a lower leakage current, although in the case of an electrolytic capacitor the output voltage would not vary much either. Considering a capacitor with a nominal voltage of 10 V, leakage currents are usually around $I_{leak} = 0.01 C V_{nom} \approx 2.2 \mu A$. This current loss in R_3 represents 0.484 mV, i.e., a 0.012% variation.

Since C_{out} is not greater than 50 μF , a protection diode between the output and input can be omitted. However, because C_{adj} is greater than 1 μF , one must be placed between ADJ (anode) and OUT (cathode) (see Figure 29)¹³.

that is within the limits of 33 m (typ) to 2.2 is required. The capacitor type can be ceramic, tantalum, or aluminum electrolytic as long as it meets the minimum capacitance value and ESR limits over the circuit's entire operating temperature range. Higher values of output capacitance can be used to enhance loop stability and transient response with the additional benefit of reducing output noise (NCP1117/D, p.9)

¹⁰Reference Voltage, Adjustable Output Devices: 1.225 V (min), 1.25 (typ), 1.270 (max); Adjustment Pin Current ($V_{in} = 11.25 V$, $I_{out} = 800 \text{ mA}$): 52 μA (typ), 120 μA (max) (NCP1117/D, Table Electrical Characteristics, p.3-4)

¹¹The value of R_1 [R_4 in the schematic] can be selected to provide the minimum required load current to maintain regulation and is usually in the range of 100 Ω to 200 Ω (NCP1117/D, p.9)

¹²The output ripple will increase linearly for fixed and adjustable devices as the ratio of output voltage to the reference voltage increases. The loss of ripple rejection can be restored with the addition of bypass capacitor C_{adj} . The reactance of C_{adj} at the ripple frequency must be less than the resistance of R_1

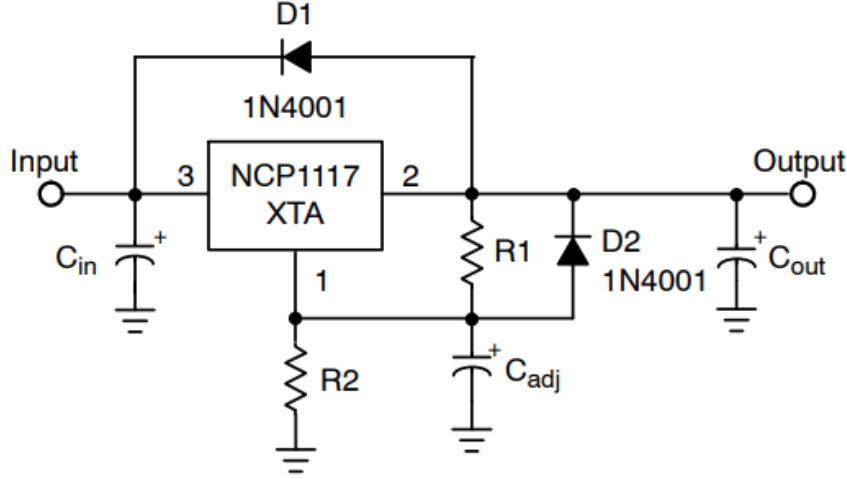


Figure 29: Protection Diode Placement (fig. 25, NCP1117)

4.2.4 Detector

The detector block consists of a Toshiba TCD1304DG linear sensor, selected mainly because of its low cost and proven use in spectroscopy prototypes, and an MCP9808T temperature sensor, available in the laboratory, used in the control of heat dissipation through a Peltier module.

For the CCD sensor, the control signals coming from the microcontroller (ICG, fM, and SH) were connected directly to the chip inputs. Unlike the datasheet (see Figure 30), a ferrite core was placed between VTCD and VTCD_A, instead of shorting them together, in order to separate the logic supply from the analog supply. An inverter was omitted because this is resolved by the program logic. Both inputs have reservoir and filtering capacitors; in particular, the one added at VTCD is 10 $\mu\text{F}/25\text{ V}$, as indicated in the aforementioned schematic.

At the output stage, a resistive divider was placed to reduce the maximum output voltage¹⁴ from 3.5 V to the 3.3 V accepted by the microcontroller¹⁵, together with a voltage follower whose function is to decouple the high output impedance of the resistive divider from the capacitive load of the ADC. Since the sampling process involves charging the ADC's internal capacitor during a limited time window, the buffer ensures a low drive impedance and favors proper voltage settling before each conversion. This operational amplifier is connected to the 4 V VTCD_A supply¹⁶ together with a capacitor bank of 10

(NCP1117/D, p.9)

¹³The NCP1117 family has two internal low impedance diode paths that normally do not require protection when used in the typical regulator applications. The first path connects between V_{out} and V_{in}, and it can withstand a peak surge current of about 15 A. Normal cycling of V_{in} cannot generate a current surge of this magnitude. Only when V_{in} is shorted or crowbarred to ground and C_{out} is greater than 50 μF , it becomes possible for device damage to occur. Under these conditions, diode D1 is required to protect the device. The second path connects between C_{adj} and V_{out}, and it can withstand a peak surge current of about 150 mA. Protection diode D2 is required if the output is shorted or crowbarred to ground and C_{adj} is greater than 1.0 μF (NCP1117/D, p.10)

¹⁴DC output signal voltage V_{os} 1.5 V (min), 2.5 V (typ), 3.5 V (max) (AKZ00038-3, p. 3)

¹⁵R6 = 1k, R7 = 10k $\rightarrow V'_{OS}(\text{max}) = \frac{10\text{ k}\Omega}{10\text{ k}\Omega + 1\text{ k}\Omega} 3.5\text{ V} = 3.182\text{ V}$

¹⁶Supply Voltage (VDD): 1.8 V (min), 6.0 V (max) (MCP6001/1R/1U/2/4, DC Electrical Specifications, p. 3); The output voltage range of the MCP6001/2/4 op amps is VDD – 25 mV (minimum), and VSS

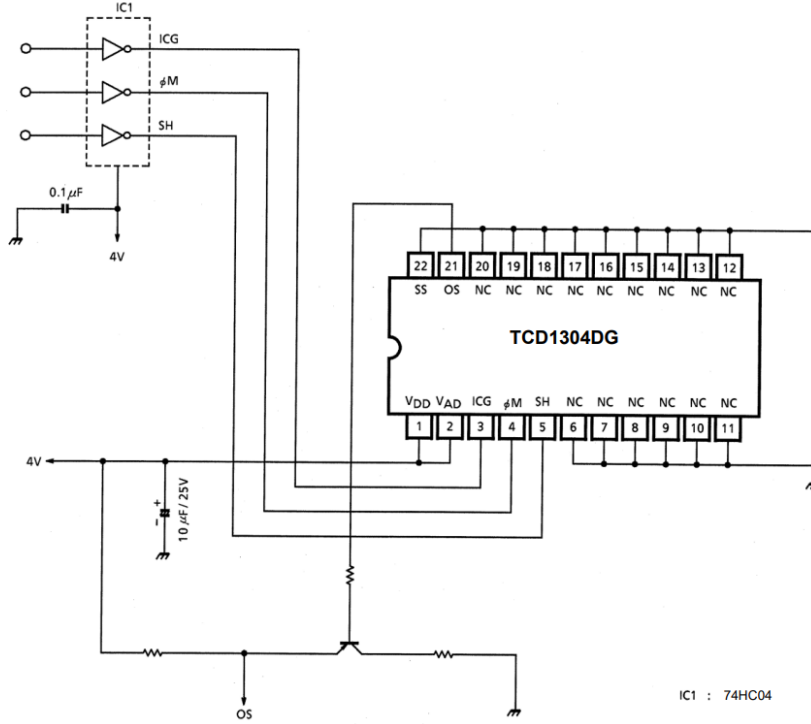


Figure 30: Typical Drive Circuit (AKZ00038-3, p. 11)

μF and 100 nF.

Figure 32 presents the typical ADC connection diagram. Considering an approximate parasitic capacitance of 5 pF and the STM32F446 internal sampling capacitor of 4-7 pF¹⁷, the total load is

$$C_L = C_p + C_{ADC} = 5 \text{ pF} + 7 \text{ pF} = 12 \text{ pF}$$

Since the operational-amplifier datasheet indicates that adding a small resistor at the output is necessary to improve stability when capacitive loads are large (greater than 100 pF in follower mode)¹⁸, it is not necessary to add it in this case (see Figure 31), also favoring fast charging of the sampling capacitor during the configured cycles.

The inclusion of an external filtering capacitor (C_{ext}) at the ADC input was analyzed. However, since the TCD1304 sensor operates at a pixel frequency of 500 kHz, maintaining the integrity of the video-signal edges was prioritized. A large-value capacitor would act as a charge reservoir, but it would also limit bandwidth, smoothing transitions between pixels and degrading the spectrometer resolution. Considering that the total pin capacitance (~ 12 pF) is significantly below the 100 pF stability limit of the MCP6001 buffer, it was decided to omit this component to ensure proper signal settling within the configured sampling cycles. In any case, pads are provided both for the RISO resistor (in connected-jumper format) and for a capacitor to ground.

+ 25 mV (maximum) when $R_L = 10 \text{ k}\Omega$ is connected to $V_{DD}/2$ and $V_{DD} = 5.5\text{V}$. (MCP6001/1R/1U/2/4, p. 13)

¹⁷CADC (internal sample and hold capacitor): 4 pF (typ), 7 pF (max) (STM32F446xC/E datasheet, Table 74)

¹⁸When driving large capacitive loads with these op amps (e.g., $> 100 \text{ pF}$ when $G=+1$), a small series resistor at the output improves the feedback loop's phase margin (stability) by making the output load resistive at higher frequencies (MCP6001, p. 14)

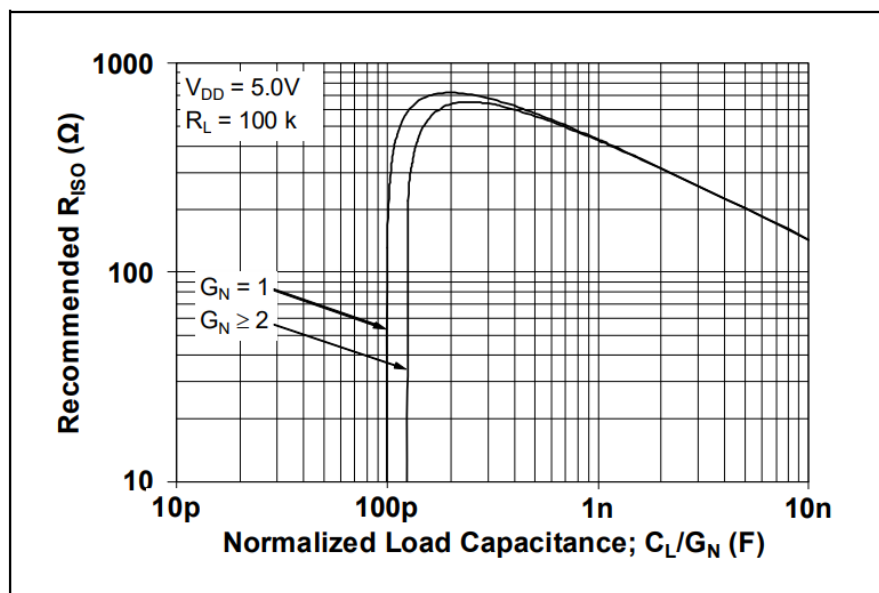


Figure 31: Recommended RISO values for capacitive loads (MCP6001, figure 4-4)

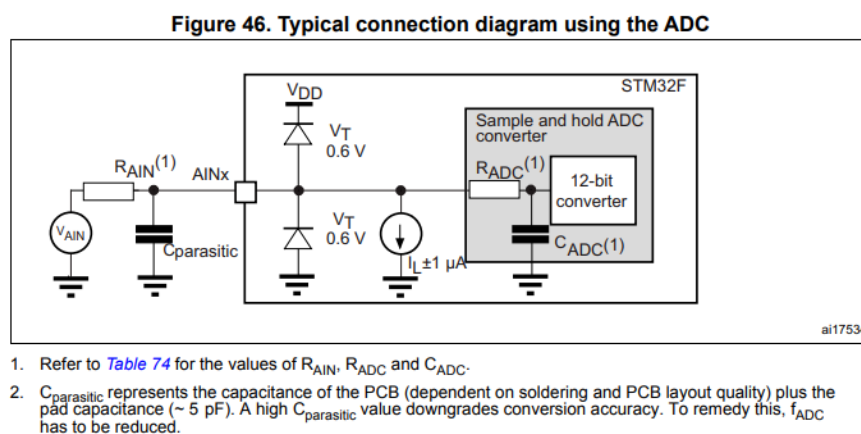


Figure 32: Typical connection diagram using the ADC (STM32F446xC/E, fig. 46)

In order to improve the performance of the TCD1304, it is proposed to sense the chip temperature using an MCP9808T-E/MC and place a Peltier module between the PCB and the CCD sensor (see PCB B section). Making use of the distance provided by the sensor's female connectors (in THT format), a metal prism (approximately 6 mm x 7.5 mm x 42 mm) would be placed with one surface in contact with the detector and another with the temperature sensor.

The MCP9808 is a digital temperature sensor with an I2C interface, offering a resolution of up to 0.0625 °C and programmable alert limits. In this project, it is powered from VDD (3.3 V)¹⁹ and its I2C address, configured through pins A0, A1, and A2, which were connected to ground, is 0x18²⁰. A 10 kΩ pull-up resistor was placed on the alert pin,

¹⁹Operating Voltage Range (VDD): 2.7 V (min), 5.5 V (max) (MCP9808, Temperature Sensor DC Characteristics, p.3)

²⁰The address pins correspond to the Least Significant bits of the address bits: A6-A5-A4-A3-A2-A1-A0: 0011xxx

since it is an open-drain output²¹.

Finally, the exposed thermal pad was connected to ground, according to the datasheet instructions²²

4.2.5 Outputs

Finally, to facilitate interaction and testing of the board, the following peripherals were included:

- An LED connected to pin PB7 and a push button on pin PB6 are provided
- Specific outputs were routed on the PCB for pins PC5, PB1, and PB10 for unrestricted use. In particular, pin PB10 provides access to channel 3 of TIM2, allowing high-precision timing functions (32-bit) to be integrated without modifying the base design.
- Test points were brought out for SH, ICG, fM, and the CCD sensor output after the operational amplifier
- The board can communicate with the microcontroller through two UART peripherals (one for data transmission and another for programming), I2C, SPI, and SWD

4.3 PCB B

Board B integrates the complementary modules of the system: two stepper-motor controllers and the power stage intended to control a Peltier module for thermal stabilization of the detector.

The general component distribution and board design can be seen in Figure 33.

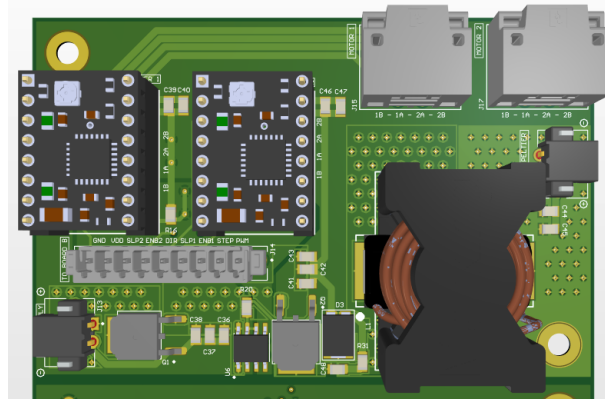


Figure 33: 3D view of PCB B, intended for control of the stepper motors and the Peltier module.

²¹The polarity [of the alert output] can also be specified as active-high or active-low using bit 1 of CONFIG. This is an open-drain output and requires a pull-up resistor (MCP9808, p. 30)

²²There is an internal electrical connections between the Exposed Thermal Pad (EP) and the GND pin. The EP may be connected to the system ground on the Printed Circuit Board (PCB) (MCP9808, p. 11)

4.3.1 Motors

For the eventual positioning of optical elements, such as the diffraction grating, or manipulation of the sample along two axes (allowing measurements at different depths or points), “BOARD B” incorporates two A4988 controllers. The use of these power drivers, designed for bipolar stepper motors, has the following characteristics:

- The **MSx** configuration pins were set high by default, which provides a resolution of 1/16. However, the PCB allows jumpers to be cut to set these values to 0, allowing the motor resolution to be varied according to the displacement-precision requirements.
- Both drivers share the STEP and DIR signals to simplify the control logic
- Each motor has its own ENABLE and SLEEP lines, allowing activation or power saving of each axis completely independently

Table 6 details the pins assigned to this stage:

Signal	Pin	Function
NSLEEP1 / NENABLE1	PB4 / PB5	Motor 1 power and activation control
NSLEEP2 / NENABLE2	PC11 / PC12	Motor 2 power and activation control
DIR	PD2	Direction control (shared)
STEP	PB9	Step generation (TIM4_CH4)

Table 6: Pin assignment for stepper-motor control.

4.3.2 Peltier module

The design includes a control loop to stabilize the temperature of the CCD sensor using a Peltier cell, with the aim of reducing the sensor dark current and stabilizing the spectrometer baseline. For this purpose, the incorporation of a Same Sky CP60140 thermoelectric module (TEC) thermally coupled to the CCD is proposed.

Power control is performed through a PWM signal generated by pin PB8 (TIM10CH1), which, through a buck-type supply, precisely regulates heat flow by varying the duty cycle. System feedback is managed by reading the temperature sensor over I2C. In addition, configurable alert interrupts from this sensor are received through pin PC2.

This module has the following technical characteristics:

- Dimensions: 15 mm x 15 mm
- Maximum voltage at maximum ΔT and $T_h = 27^\circ C$: $2.1 V_{dc}$
- Maximum input current to reach maximum ΔT : 6 A
- Typical internal resistance measured at $25^\circ C$: $0.29 \Omega (\pm 10\%)$
- Maximum heat transfer (Q_{max}):

$$- 7.1 \text{ W } (T_h = 27^\circ C)$$

– 7.9 W ($T_h = 50^\circ\text{C}$)

Maximum heat absorption on the cold side occurs at $I_{\max}$, $V_{\max}$, and $\Delta T = 0^\circ\text{C}$

- Maximum temperature difference ($\Delta T_{\max}$);

– 68°C ($T_h = 27^\circ\text{C}$)

– 75°C ($T_h = 50^\circ\text{C}$)

The maximum temperature difference occurs at $I_{\max}$, $V_{\max}$, and $Q = 0\text{ W}$ ($\Delta T_{\max}$ measured under vacuum at 1.3 Pa)

Initially, the use of a chopper-type switching controller (such as the A4988) or the direct application of a PWM signal to drive the thermoelectric module (TEC) was evaluated, although it was ultimately discarded. According to a technical compendium prepared by the specialized firm Meerstetter Engineering²³ on industry design guidelines, low-ripple direct-current (DC) power is a fundamental requirement for operating these devices. In this technical document, Meerstetter compiles warnings from the main manufacturers of thermoelectric semiconductors. It highlights that companies such as Ferrotec and Marlow require the power-supply ripple to be limited to a maximum of 10% (preferably below 5%) and explicitly advise against ON/OFF control topologies. Likewise, citing the firm RMT, the report emphasizes that direct PWM control is inherently less effective than pure DC current, since abrupt transients generate parasitic heat through the Joule Effect (I^2R), which compromises the maximum cooling capability (ΔT).

For this reason, following the guidelines of the Texas Instruments Application Report “Basic Calculation of a Buck Converter’s Power Stage” (SLVA477B), a buck converter was implemented. The same supply used for the motors, intended to be 12 V, is used as the V_{in} power source.

Duty cycle. The maximum duty cycle is calculated as²⁴:

$$D = \frac{V_{\text{out}}}{V_{\text{in}} \times \eta} = \frac{2.1\text{ V}}{12\text{ V} \times 0.9} = 0.194...$$

Input capacitor. An input capacitor²⁵ large enough to stabilize the input voltage due to the current peak required by switching is needed. Ceramic capacitors with low equivalent series resistance (ESR), using X5R dielectric or better, are preferred. The value may be increased if the input source is noisy. In this case, a 100 μF electrolytic capacitor is proposed as a reservoir and a 10 μF X7R ceramic capacitor for high-frequency filtering.

Switch. The **AOD417 P-Channel MOSFET**, available in the laboratory, was selected. Unlike an N-channel device, using a P-channel device on the high side of the buck allows the Source to be referenced to the input voltage ($V_{BB2} = 12\text{ V}$), simplifying gate drive, since turning on the component only requires bringing the Gate to a potential lower than the Source.

The AOD417 has the following characteristics²⁶:

²³<https://www.meerstetter.ch/customer-center/compendium/71-peltier-element-efficiency>

²⁴SLVA477B, Eq. 1, p. 1

²⁵SLVA477B, sec. 6, p. 4

²⁶AOD417, Features, p. 1

- Drain-Source voltage: $V_{DS} = -30\text{ V}$, providing a 150% safety margin with respect to the 12 V input
- Drain current: $I_D(\text{max}) = -25\text{ A}$, exceeding the 6 A required by the Peltier cell
- On resistance: $R_{DS(\text{ON}, \text{max})} = 50\text{ m}\Omega$ @ $V_{GS} = -10\text{ V}$, where the worst-case scenario was considered (upper limit of the resistance value at $V_{GS} = -10\text{ V}$ at room temperature) and the highest temperature (150°C) (factor ~ 1.46 , AOD417, fig. 4, p. 3)
- $t_r = 30\text{ ns}$
- $t_f = 26\text{ ns}$
- $R_{\theta JA}(\text{max}) = 50^\circ\text{C/W}$
- $C_{iss} = 920\text{ pF}$

Switching frequency To find the maximum switching-frequency limit, the minimum charging time of the transistor capacitor must be known, and therefore the gate resistance must first be determined. A large gate resistance reduces the switching speed, resulting in increased power loss, reduced performance, and potential heating problems. A low value, on the other hand, increases the switching speed but can cause voltage increases and oscillations that may lead to device failure²⁷. The following are considered for the calculation

- MOSFET
 - When the switching node is brought to 0 V, $V_{DS} = 12\text{ V}$
 - $R_{DS(\text{on}), \text{typ}} = 36\text{ m}\Omega$
 - $Q_G = 16.2\text{ nC}$
 - $R_g(\text{typ}) = 6\text{ }\Omega$
- DRIVER
 - $I_{\text{peak}} = 6\text{ A}$
 - $V_{\text{out}} = 25\text{ mV} \sim (V_{DD} - 25\text{ mV})$
 - V_{DD}

A minimum resistance equal to

$$R_g = \frac{V_{DD}}{I_{\text{peak}}} = 2\text{ }\Omega$$

is recommended.

It is therefore decided to place a $10\text{ }\Omega$ resistor. In this way, the C_{iss} capacitor charges in approximately

$$5\tau = 5 R_g C_{iss} = 5 \cdot 16\text{ }\Omega \cdot 920\text{ pF} = 73.6\text{ ns}$$

²⁷RENESAS knowledge base

Therefore, one transition (composed of two edges) takes $0.1472 \mu s$. Applying a margin of $T_{\text{switching}} = 0.2 T_{\text{open}}$, the minimum required switching period is

$$T_{\text{conm}}(\text{min}) = \frac{0.1472 \mu s}{0.2} = 0.736 \mu s$$

Consequently, the maximum switching frequency is

$$f_{\text{sw}}(\text{max}) = \frac{1}{0.736 \mu s} = 1358.69 \text{ kHz}$$

For the minimum frequency, the audible-range value, i.e., 20 kHz, was used.

There is a direct trade-off between dynamic performance and converter efficiency. While frequencies in the 1 MHz range allow the size of the inductor and capacitor to be minimized, they increase thermal stress in the semiconductors, reducing efficiency. In a Buck converter, there are three dominant mechanisms of power loss affected by frequency²⁸:

- Switching Losses: these are associated with the switch transition between on and off states. Since the number of transitions per second increases with frequency, these losses are directly proportional to f_{sw}
- Driver Losses: the power dissipated by the TC4420 driver to control the gate voltage is defined as

$$P_{\text{driver}} = V_{\text{drive}} Q_g f_{\text{sw}}$$

where V_{drive} is the drive voltage (12 V) and Q_g is the total gate charge of the MOSFET. Again, this power increases linearly with frequency

- Conduction Losses: these occur when the MOSFET or diode is in the active state. They are independent of frequency and depend mainly on duty cycle and component resistance

$$\begin{aligned} P_{\text{cond, MOSFET}} &= I_{\text{RMS}}^2 R_{\text{DS(on)}} \\ P_{\text{cond, diode}} &= V_{\text{diode}} I_{\text{output}} (1 - D) \end{aligned}$$

The total power dissipated by the MOSFET is²⁹:

$$\begin{aligned} P_{\text{tot}} &= P_{\text{cond}} + P_{\text{sw}} \\ &= I_{\text{out}}^2 R_{\text{DS(ON)}} D + \frac{1}{2} V_{\text{in}} I_{\text{out}} (t_r + t_f) f_{\text{sw}} \\ &= (6A)^2 50 \text{ m}\Omega \cdot 0.194 + \frac{1}{2} 12 \text{ V} 6 \text{ A} (30 \text{ ns} + 26 \text{ ns}) f_{\text{sw}} \\ &= 0.3492 \text{ W} + 2.016 \cdot 10^{-6} \text{ Ws} \cdot f_{\text{sw}} \end{aligned}$$

With the package thermal resistance:

²⁸SLVAED3A - Application note: How the Switching Frequency Affects the Performance of Buck Converter

²⁹For the switching-power calculation, the energy lost during turn-on (triangle formed by V_{in} and I_{out} during time t_r) and the energy lost during turn-off (calculated analogously during t_f) are considered. The sum of both is multiplied by the number of times it occurs per second (f_{sw}) to obtain the average power

$$\Delta T = (0.3492 \text{ W} + 2.016 \cdot 10^{-6} \text{ Ws} \cdot f_{sw}) \cdot 50^\circ\text{C/W}$$

For the inductor calculation, the inductor current ripple ³⁰ is calculated as:

$$\begin{aligned} \Delta I_L &= \frac{(V_{\text{IN(max)}} - V_{\text{OUT}}) \times D}{f_s \times L} \\ &= \frac{(12 \text{ V} - 2.1 \text{ V}) \times 0.1944}{f_s \times L} = \frac{1.92456 \text{ V}}{f_s L} \end{aligned}$$

As mentioned previously, the ripple must be less than 10% and, preferably, less than 5%. With a current of 6 A, this is:

$$\begin{aligned} \Delta I_L &= \frac{1.92456 \text{ V}}{f_s L} < 5\% \cdot 6 \text{ A} = 0.3 \text{ A} \\ \rightarrow L &> \frac{1.92456 \text{ V}}{f_s \cdot 0.3 \text{ A}} = 6.4152/f_s [\text{H}] \end{aligned}$$

For the output capacitor, one with the same characteristics as the input capacitor is required in order to minimize output-voltage ripple. The value can be calculated as ³¹:

$$C_{\text{OUT(min)}} = \frac{\Delta I_L}{8 \times f_s \times \Delta V_{\text{OUT}}} = \frac{0.3 \text{ A}}{8 \times f_s \times 0.021 \text{ V}} = 1.79/f_s [\text{F}]$$

where a voltage ripple ΔV_{OUT} equal to 1% of the value used in the Peltier (2.1 V) was used. The selected value for this capacitor is 47 μF .

For comparison at different frequency values, Table 7 is presented.

f_{sw}	L(min)	C(out, min)	P_{MOSFET}	T_j	$(t_r + t_f)/t_{sw}$
50 kHz	128.4 μH	35.8 μF	0.45 W	47.5 $^\circ\text{C}$	0.28%
100 kHz	64.16 μH	17.9 μF	0.55 W	52.5 $^\circ\text{C}$	0.56%
200 kHz	32.08 μH	8.95 μF	0.75 W	62.5 $^\circ\text{C}$	1.12%
300 kHz	21.39 μH	5.97 μF	0.954 W	72.7 $^\circ\text{C}$	1.68%
400 kHz	16.04 μH	4.48 μF	1.1556 W	82.78 $^\circ\text{C}$	2.24%
500 kHz	12.84 μH	3.58 μF	1.36 W	93 $^\circ\text{C}$	2.8%
1 MHz	6.42 μH	1.79 μF	2.37 W	143.5 $^\circ\text{C}$	5.6%

Table 7: Comparison of characteristics for different switching frequencies

The selected model must be suitable in terms of saturation current (I_{SAT}) to avoid inductance collapse, power losses due to DC resistance (R_{DC}), and cost. Based on the list of inductors available in the laboratory, the **CODACA CSBX1275-100M** could be used for 500 kHz or the **Würth 74437529203220** for 300 kHz. Considering that operating with the second option provides a 58% margin with respect to the silicon limit

³⁰SLVA477B, Eq. 2, p. 2

³¹SLVA477B, eq. 12, p. 4

and a 51.53% margin with respect to the datasheet-table limit, and that the selected inductor has a saturation current of 21.7 A, this switching frequency is selected. Some details³²:

- **Thermal and Power Efficiency:** The model has a significantly low DC resistance (R_{DC}), with a maximum of 5.39 m Ω , generating much lower heat dissipation than, for example, the Würth model selected for the 50 kHz frequency (150 μ H, 30.6 m Ω)
- **Current Margin:** The selected model has a maximum rated current $I_{rated} = 23.9$ A and typical saturation currents of $I_{SAT,10\%} = 7.5$ A, $I_{SAT,30\%} = 21.7$ A. Since the Peltier requires a constant current of up to 6 A, this component guarantees “cool” operation for extended periods
- **Frequency Response (f_{res}):** The model has a self-resonant frequency of 9 MHz, well above those used.

Driver. To drive this MOSFET, the TC4420COA driver is used. This integrated circuit³³:

- Operates with logic voltages, since the minimum high input is 2.4 V
- Supports a supply voltage between 4.5 and 18 V
- Can deliver current peaks of up to 6 A³⁴
- Considering that the total gate charge at 10 V of the AOD417³⁵ is 16.2 nC, the driver must provide twice³⁶ the stored energy $E = QV/2$, i.e.:

$$E = QV \stackrel{V_S(\text{init})=0}{=} Q_G V_{BB} = 16.2 \text{ nC} \cdot 12 \text{ V} = 194.4 \text{ nJ}$$

In this way, the applied power is

$$P = \frac{\text{Energy}}{1 \text{ cycle}} \cdot \frac{\text{cycles}}{1 \text{ s}} = E \cdot f_{sw} = 194.4 \text{ nJ} \cdot 300 \text{ kHz} = 58.32 \text{ mW}$$

Since the datasheet indicates a maximum quiescent current I_S of 3 mA for the TC4420³⁷, the total power is:

$$P_{\text{total}} = 58.32 \text{ mW} + 3 \text{ mA} \cdot 12 \text{ V} = 94.32 \text{ mW}$$

It follows from the last point that the value is lower than the 470 mW dissipation capability of the 8-SOIC footprint at an ambient temperature below 70°C³⁸.

Considering the previously obtained ripple ΔI_L , the maximum switching current of the system is calculated as³⁹:

³²74437529203220, Electrical Properties, p. 1

³³TC4420, DC CHARACTERISTICS (OVER OPERATING TEMPERATURE RANGE), p.4

³⁴TC4420, Features, p. 1

³⁵AOD417, Electrical Characteristics, p. 2

³⁶<http://hyperphysics.phy-astr.gsu.edu/hbase/electric/capeng2.html>

³⁷TC4420, Power Supply Current, DC CHARACTERISTICS (OVER OPERATING TEMPERATURE RANGE), p. 4

³⁸TC4420, Absolute Maximum Ratings, p. 3

³⁹SLVA477B, Eq. 4, p. 2

$$\begin{aligned}
I_{\text{SW(max)}} &= \frac{\Delta I_L}{2} + I_{\text{OUT(max)}} \\
&= \frac{0.3 \text{ A}}{2} + 6 \text{ A} \\
&= 6.15 \text{ A}
\end{aligned}$$

It should be noted that the current limit of the selected MOSFET ($|I_D(\text{max})| = 25 \text{ A}$) is well above this value.

Diode. For the diode, a Schottky diode is selected in order to reduce losses⁴⁰. The forward current rating must be⁴¹:

$$\begin{aligned}
I_f &= I_{\text{out(max necessary)}} \times (1 - D) \\
&= 6 \text{ A} (1 - 0.194) = 4.836 \text{ A}
\end{aligned}$$

In addition, the power dissipation of the diode must be checked, since it must be able to withstand

$$P_D = I_f \times V_f$$

The **Onsemi FSV15150VTR-ND**, available in the laboratory, meets these requirements, since it has:

- $I_f = 15 \text{ A} > 4.836 \text{ A}$
- $V_{f, 5\text{A-}100^\circ\text{C}} \approx 0.6 \text{ V} \rightarrow P_D = 0.6 \text{ V} \times 4.836 \text{ A} = 2.91 \text{ W}$ ⁴²
- It has an operating temperature range of up to 150° at $T_A = 25^\circ\text{C}$
- Applying a 30 mm x 30 mm square pad, the junction-to-ambient thermal resistance is 40°C/W . For a dissipation of 2.91 W , the junction-temperature rise relative to ambient is:

$$\Delta T_J = R_{\theta JA} P_D = 40^\circ\text{C/W} \times 2.91 \text{ W} = 116.4^\circ\text{C}$$

Considering an ambient temperature of 25°C , the estimated junction temperature is:

$$T_J = T_A + \Delta T_J = 25^\circ\text{C} + 116.4^\circ\text{C} = 141.4^\circ\text{C} < 150^\circ\text{C}$$

⁴⁰SLVA477B, sec. 4, p. 3

⁴¹SLVA477B, eq. 7, p. 3

⁴²SLVA477B, fig. 3, p. 3

5. Signal processing

A measured Raman spectrum contains, in addition to the useful signal, different noise and background contributions that can make identification of the spectral bands difficult. Processing aims to reduce these disturbances without artificially modifying the position, shape, or relative intensity of the signals of interest.

Pelletier’s work [2], which presents a review of noise sources and preprocessing techniques used in Raman spectroscopy, was used as a general reference for this stage. This reference is used as an introductory framework and is not intended to constitute an exhaustive literature review of each implemented technique.

5.1 Noise sources

The main unwanted contributions present in a Raman measurement include:

- **Photon *shot noise*.** It is due to the statistical nature of photon detection. Its standard deviation increases approximately with the square root of the number of detected photons. It is an unavoidable source of noise and, when it dominates the measurement, improving the signal-to-noise ratio requires increasing the number of detected photons.
- **Sample-generated noise.** This includes variations associated with temperature, heterogeneity, bubbles, or particles, as well as non-Raman optical contributions. One of the most important is fluorescence, which usually appears as a slowly varying baseline and can be considerably more intense than the Raman bands.
- **Instrumental noise.** This includes thermal and detector readout noise, fixed pattern, pixel-to-pixel variations, calibration drift, and changes in optical coupling to the sample. These effects can modify both the absolute level and the repeatability of the spectra.
- **Processing-generated noise.** The mathematical operations applied to the spectrum can themselves introduce artifacts. An incorrect *dark* subtraction, smoothing operation, or *baseline* estimate may alter the useful signal or generate artificial variations.
- **External noise.** This may come from ambient lighting or other optical sources external to the system. Very intense isolated events may also occur, such as those associated with cosmic rays, affecting only one or a few detector elements.

5.2 Raman spectrum preprocessing

The different noise sources have different characteristics and can therefore be addressed using specific tools. Random noise can be reduced by repeated measurements and smoothing; isolated events can be identified from their narrow spatial width; and slowly varying contributions, such as fluorescence, can be estimated using *baseline* correction techniques.

Repeatedly measuring the same sample is particularly useful. When N equivalent spectra are summed, the repeatable signal increases proportionally to N , while uncorrelated random noise increases approximately as $\sqrt{N}$. Equivalently, when acquisitions are averaged, the mean signal is preserved while the noise deviation decreases approximately as:

$$\sigma_{\text{average}} = \frac{\sigma}{\sqrt{N}}$$

so the signal-to-noise ratio improves approximately as $\sqrt{N}$ [2].

Smoothing can be used to reduce high-frequency components, although it must be applied moderately, since excessive filtering can broaden the spectral bands and reduce the effective resolution.

On the other hand, a slowly varying *baseline* can reduce the contrast between Raman bands and the background. A greater *baseline* intensity in a region does not necessarily imply a greater intrinsic intensity of the Raman bands present in that region; however, it can make weak signals more difficult to identify. Its estimation and subtraction can therefore facilitate interpretation of the spectrum.

These considerations were used to define the processing procedure implemented in the software.

5.3 Proposed processing procedure

The procedure is defined as a sequential chain applied to the acquired spectra. The initial objective is not to perform absolute quantification, but to obtain corrected and comparable signals that make it possible to distinguish potential Raman bands from noise, *baseline*, and artifacts.

The different operations were implemented in a configurable manner, making it possible to evaluate both the original data and the individual effect of each processing stage.

5.3.1 Dark-signal subtraction

For a given acquisition configuration, a dark reference, or *dark*, can be obtained by blocking illumination on the detector.

If $S_{\text{raw}}(i)$ represents the raw spectrum and $D(i)$ the dark signal corresponding to pixel i , a conventional correction can be expressed as:

$$S_1(i) = S_{\text{raw}}(i) - D(i)$$

Since a single dark acquisition also contains random noise, multiple measurements are used and their average is obtained:

$$D(i) = \frac{1}{N} \sum_{j=1}^N D_j(i)$$

In the developed system, in addition, the CCD output has an inverse polarity with respect to optical intensity. For this reason, the software allows the acquired counts to be displayed directly or the signal to be inverted. When the dark reference and the optical-intensity representation are used simultaneously, the correction is implemented as:

$$S_1(i) = D(i) - S_{\text{raw}}(i)$$

with negative values subsequently limited to zero.

5.3.2 Detection and correction of *spikes*

After the initial correction, very narrow and intense isolated events are searched for. These may be associated with cosmic rays, anomalous pixels, or other artifacts. Unlike a real Raman band, they usually affect only one or a few pixels.

The algorithm compares each point with the median of a local window:

$$M(i) = \text{median}\{S(i - k), \dots, S(i + k)\}$$

and defines the residual:

$$r(i) = S(i) - M(i)$$

To obtain a robust measure of local dispersion, the median absolute deviation, MAD, is used:

$$\sigma_r \approx 1.4826 \text{ MAD}(r)$$

In the implementation, this estimate is obtained locally over the signal residuals. A point is marked as a *spike* when:

$$|r(i)| > T \sigma_r$$

where T is a configurable factor.

For example, for the window:

$$[100, 102, 101, 180, 103, 101, 102]$$

the local median is approximately 102, so for the central point:

$$r(i) = 180 - 102 = 78$$

If the local estimate is:

$$\text{MAD} = 1$$

then:

$$\sigma_r \approx 1.4826$$

and, for $T = 10$:

$$78 > 10 \cdot 1.4826 = 14.826$$

so the point is identified as anomalous.

The detected values are not removed, but replaced by the local median, avoiding the generation of artificial discontinuities in the spectrum.

5.3.3 Averaging of repetitions

Each spectrum is initially corrected individually using the operations enabled for a single acquisition. Subsequently, when N spectra are collected, the point-by-point average is calculated:

$$\bar{S}(i) = \frac{1}{N} \sum_{j=1}^N S_j(i)$$

The standard deviation is also calculated:

$$\sigma(i) = \sqrt{\frac{1}{N-1} \sum_{j=1}^N (S_j(i) - \bar{S}(i))^2}$$

This quantity makes it possible to estimate the repeatability of each region of the spectrum and distinguish relatively stable areas from those dominated by noise or experimental variations.

5.3.4 Smoothing

A Savitzky–Golay filter was implemented to reduce high-frequency fluctuations. This method locally fits a polynomial to a data window and allows smoothing to be performed while preserving the shape of spectral structures better than a simple moving average.

The main parameters are:

- window length w , which must be odd;
- polynomial order p , with $p < w$.

A small window produces limited smoothing, while an excessively large window can modify the width and height of the bands. For this reason, both parameters remain configurable and must be selected experimentally according to the observed resolution.

The use of mild filtering responds to the trade-off between random-noise reduction and preservation of spectral information [2].

5.3.5 Baseline correction

Fluorescence and other optical contributions can introduce a slowly varying component on which the Raman bands are superimposed. In simplified form, the measured signal can be represented as:

$$S_{\text{measured}}(i) = S_{\text{Raman}}(i) + B(i) + n(i)$$

where $B(i)$ represents the *baseline* and $n(i)$ the remaining noise contributions.

Once the background has been estimated using a function $\hat{B}(i)$, the corrected spectrum is:

$$S_{\text{corr}}(i) = S_{\text{measured}}(i) - \hat{B}(i)$$

The *asymmetrically reweighted penalized least squares* (arPLS) method was used in the developed software. This algorithm searches for a smooth baseline and iteratively

updates the weight of the different points in the spectrum, reducing the influence of positive structures associated with possible peaks.

The configurable parameters include:

- λ , which controls the smoothness of the *baseline*
- the order used in the difference penalty
- the maximum number of iterations
- the tolerance used as the convergence criterion

A high *baseline* contribution does not necessarily imply that the Raman bands present in that region have greater intrinsic intensity. However, it can reduce their contrast with respect to the background and make weak signals more difficult to identify. For this reason, the original signal is preserved and the correction can be enabled or disabled from the software.

5.3.6 Normalization

After the previous corrections, normalization can be applied to facilitate comparison of the relative shape of different spectra.

The implementation uses min–max normalization:

$$S_{\text{norm}}(i) = \frac{S(i) - S_{\min}}{S_{\max} - S_{\min}}$$

so that:

$$0 \leq S_{\text{norm}}(i) \leq 1$$

This transformation removes the absolute intensity scale and is mainly useful for qualitative comparison and visualization. For this reason, it should not be interpreted as a normalization intended directly for quantitative concentration analysis.

5.3.7 Peak detection

Finally, the software allows local maxima compatible with spectral bands to be identified.

Detection is based mainly on peak prominence. To adapt the threshold to the noise level of each spectrum, a robust estimate is first obtained from the differences between consecutive points:

$$d(i) = S(i + 1) - S(i)$$

The MAD of these differences is calculated:

$$\text{MAD}_d = \text{median}(|d - \text{median}(d)|)$$

and the characteristic noise deviation is estimated as:

$$\sigma_{\text{noise}} = \frac{\text{MAD}_d}{0.67448975\sqrt{2}}$$

The minimum required prominence is then:

$$P_{\min} = K \sigma_{\text{noise}}$$

where K is a configurable parameter.

In addition to prominence, a minimum distance between maxima and a minimum allowed width are specified. These criteria reduce the detection of small fluctuations as possible bands.

Automatic detection is only an auxiliary tool for locating spectral structures. Identification of a band subsequently requires relating its position to a spectral reference or to an appropriate calibration.

6. Acquisition and processing software on Raspberry Pi

High-level processing and the user interface were implemented in Python on Raspberry Pi. The application was developed using PySide6 for the graphical interface and `pyqtgraph` for real-time visualization of the spectra.

The software architecture was divided into modules with differentiated responsibilities:

- **SpectrometerDriver**: responsible for serial communication with the STM32, sending commands, and receiving CCD frames.
- **AcquisitionThread**: performs acquisition in a thread separate from the graphical interface, preventing UART read operations from blocking the GUI.
- **SignalProcessor**: implements the different configurable processing stages described previously.
- **RamanGUI**: concentrates user interaction, STM32 configuration, data visualization, spectral calibration, and storage and import functions.

The main program only creates the Qt application and instantiates the main window, while communication, processing, and presentation remain separated in the previous modules.

6.1 Communication and acquisition

The Raspberry Pi communicates with the STM32 through UART at 921600 baud. The **SpectrometerDriver** module implements the protocol described previously and is responsible both for sending commands and for synchronizing reception with the header of each packet.

The commands available from the interface include:

- integration-time configuration
- configuration of the number of frames to skip
- STM32 reset
- test LED toggling

Additionally, when the software runs on Raspberry Pi, an independent GPIO control is available to enable or disable the laser.

Continuous reading is performed through `AcquisitionThread`, a dedicated `QThread` that waits for new frames without blocking the main interface thread. Each valid received packet contains the 3694 samples acquired by the CCD.

For a normal acquisition, each frame generates two parallel paths:

- a copy of the original data is delivered to the interface through the `raw_data_ready` signal, allowing it to be stored without processing
- the data are sent to the `SignalProcessor` object, which applies the enabled operations and returns the spectrum to be displayed through `data_ready`

This separation makes it possible to always preserve the original information regardless of the processing configuration used during visualization.

6.2 Graphical interface

The main interface brings together instrument control and signal visualization in the same window. In the side panel, the STM32 parameters can be configured, acquisition and the laser can be controlled, a dark reference can be acquired, and the different processing stages can be enabled individually.

Figure 34 shows an example of a neon-lamp spectrum acquired with an integration time of 1 s. In this case, the signal is displayed with the polarity corresponding to optical intensity, but without yet applying *dark* subtraction.

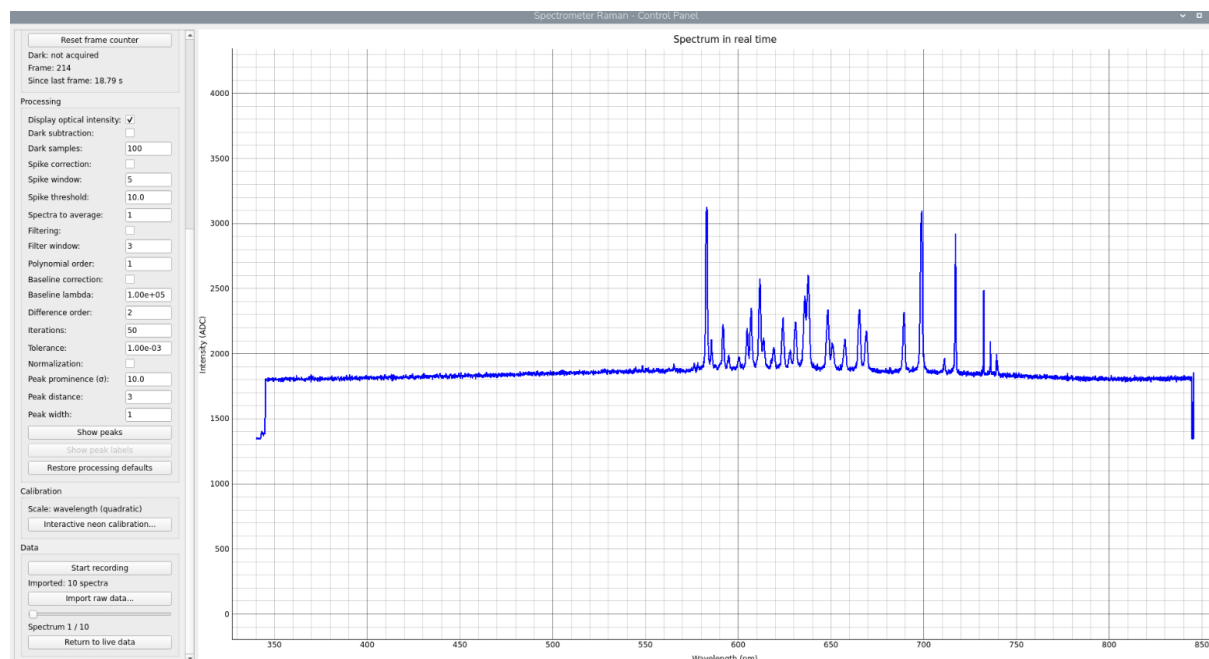


Figure 34: Neon-lamp spectrum acquired with an integration time of 1 s before applying dark-signal correction.

6.3 *Dark* acquisition and subtraction

The interface allows a dark reference to be acquired automatically using a configurable number of frames. During this operation, the received frames are stored temporarily and do not enter the normal processing pipeline.

In the configuration used for the tests, 100 dark samples were acquired with the same 1 s integration time used for the neon spectrum. Figure 35 shows the progress of this acquisition.

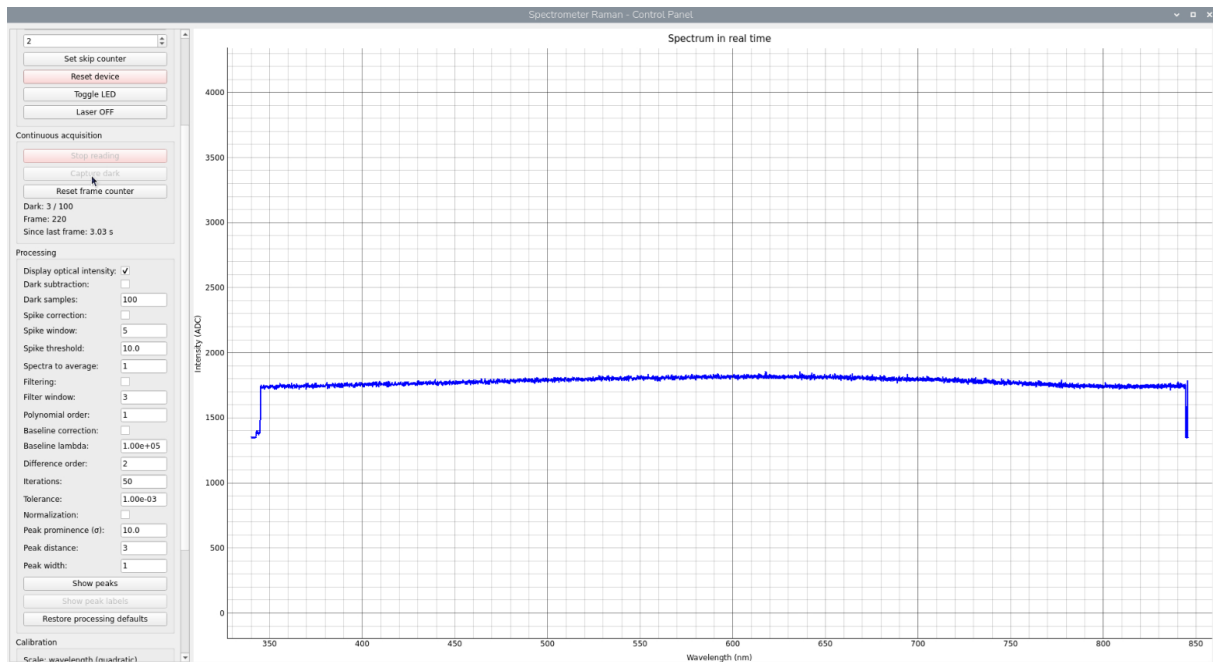


Figure 35: Acquisition process of 100 samples for dark-signal estimation.

Once the acquisitions are complete, the software calculates the point-by-point average and stores the result as a dark reference. From that point onward, the correction can be enabled or disabled from the interface itself.

Figure 36 shows the interface after acquiring the reference and enabling *dark* subtraction.

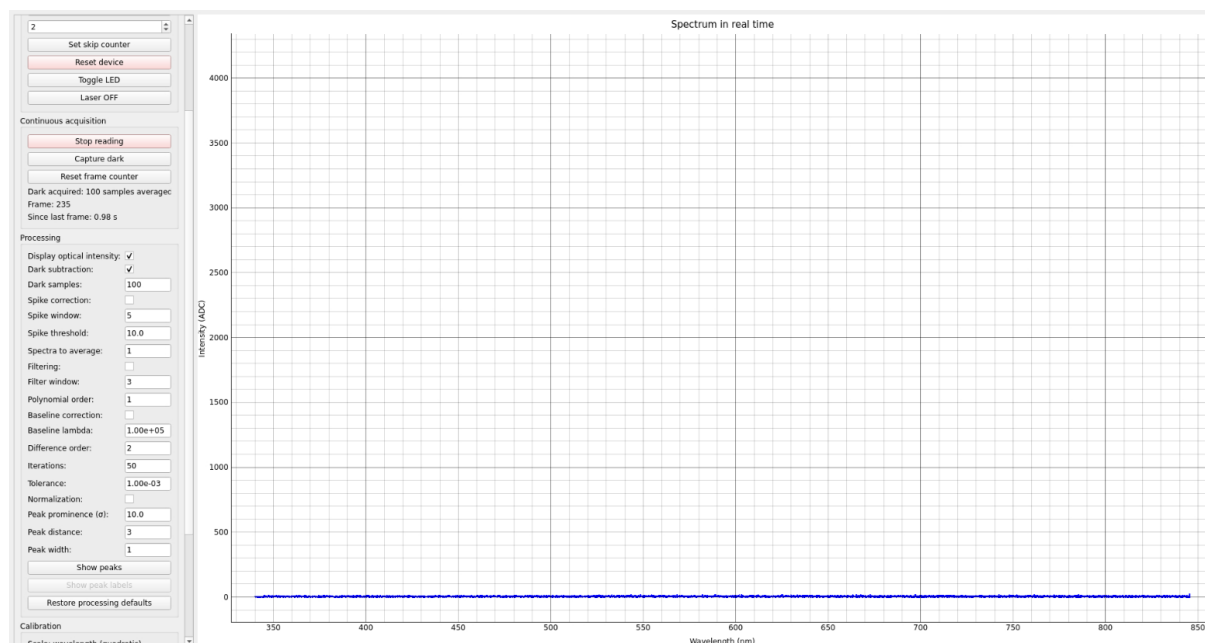


Figure 36: Interface after completing the *dark* acquisition and enabling its use.

The effect on the neon spectrum can be seen in Figure 37. Removal of the dark component shifts the background level and makes it possible to display the received optical signal more clearly.

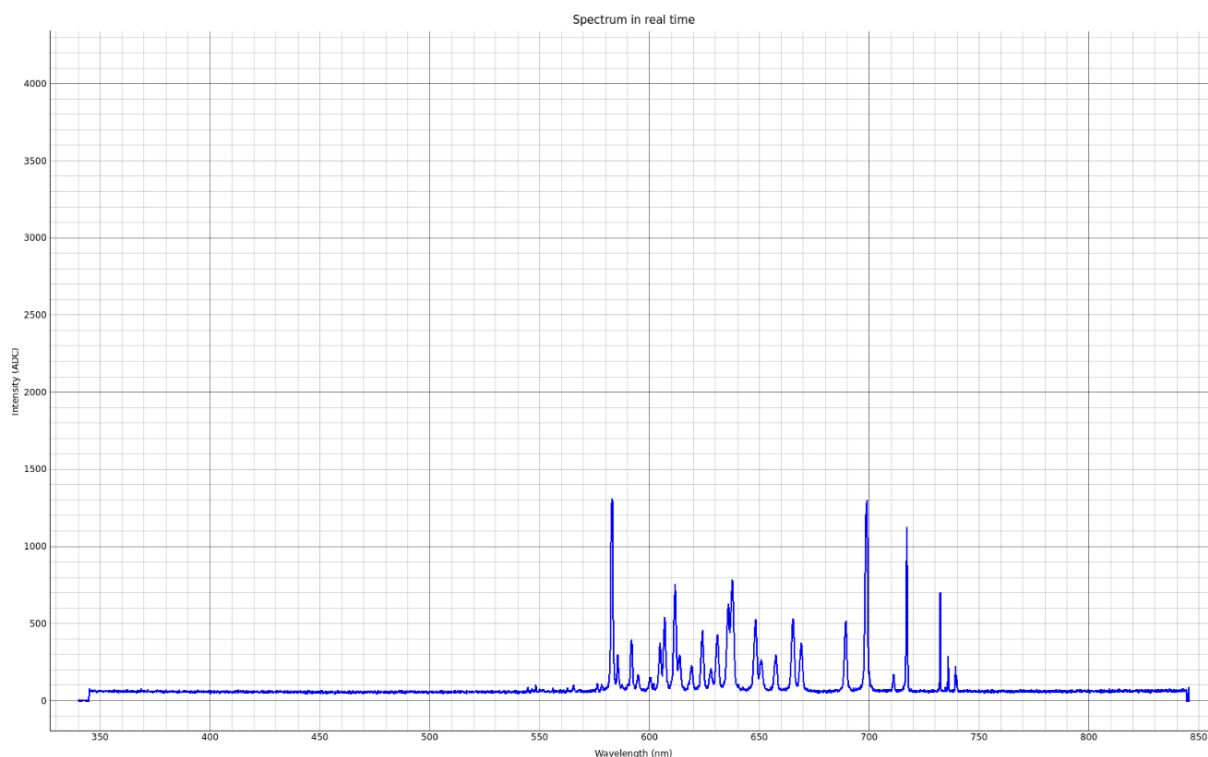


Figure 37: Neon spectrum after applying the correction using the acquired dark reference.

6.4 Interactive spectral calibration

The acquired data are initially expressed as a function of pixel index. To obtain a wavelength scale, an interactive calibration tool based on known lines from a neon lamp was implemented.

When the calibration window is opened, vertical lines corresponding to reference wavelengths are superimposed on the measured spectrum. These lines can be moved manually to the vicinity of the corresponding experimental peak.

Figure 38 shows the initial calibration window.

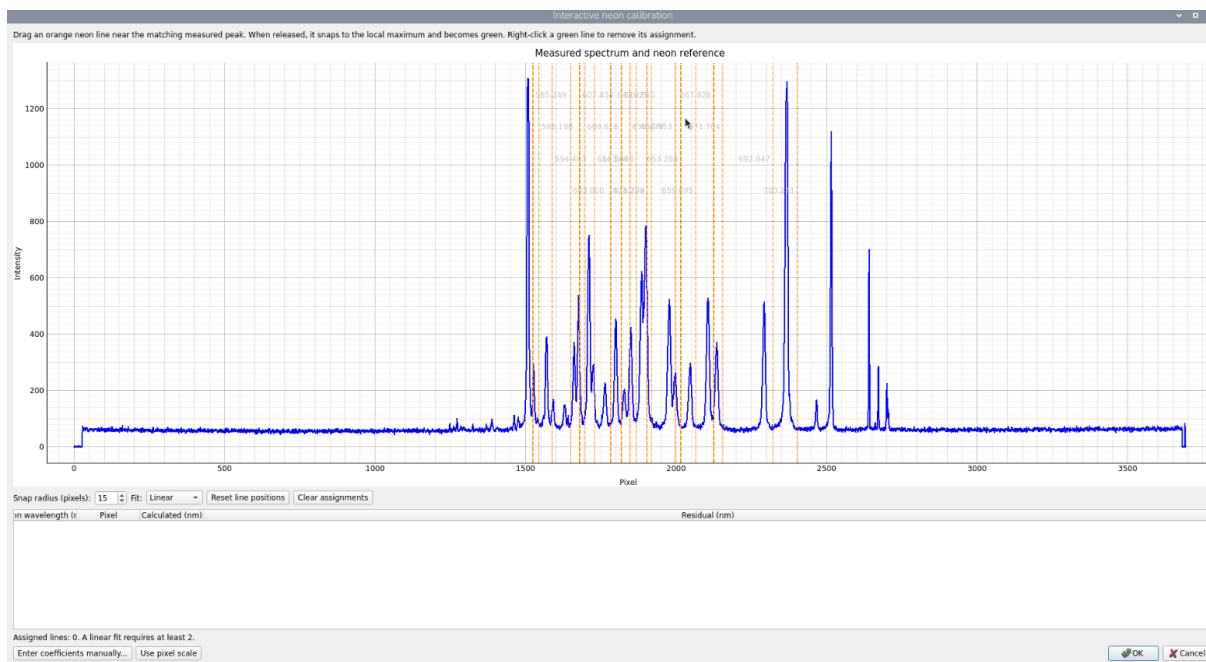


Figure 38: Interactive calibration window using reference spectral lines from a neon lamp.

When a line is moved and released, the software automatically searches for the local maximum within a configurable radius and adjusts the reference position to that pixel. Lines that have already been associated are displayed differently, as shown in Figure 39. A selected line can be unselected by right-clicking on it.

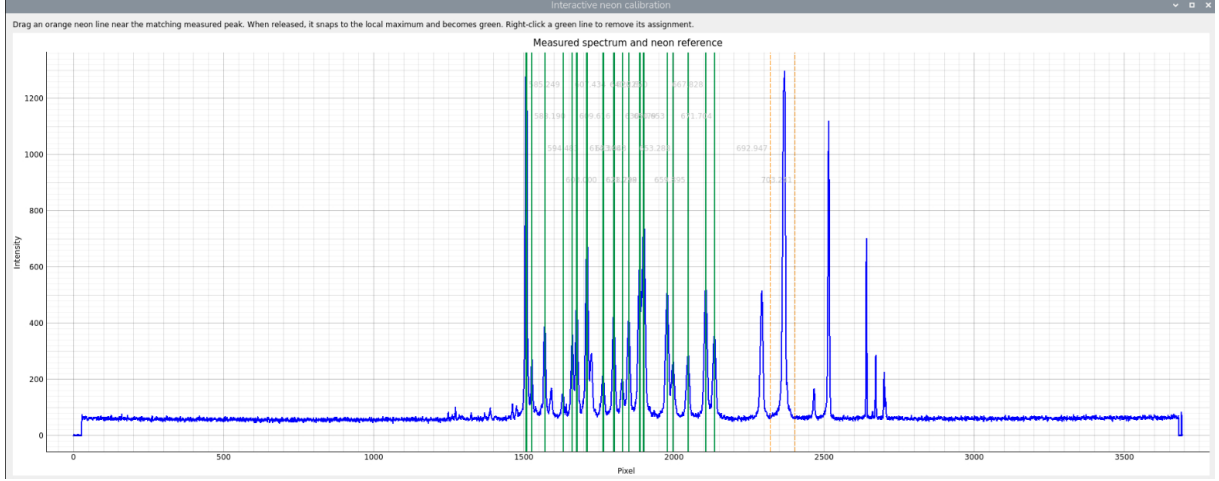


Figure 39: Assignment of neon reference lines to experimental maxima in the acquired spectrum.

From the pairs:

$$(x_i, \lambda_i)$$

where x_i represents the selected pixel position and λ_i the known wavelength of the neon line, a linear fit can be performed:

$$\lambda(x) = a_1x + a_0$$

or a quadratic fit:

$$\lambda(x) = a_2x^2 + a_1x + a_0$$

For each point, the interface displays the reference wavelength, the selected pixel, the value calculated by the model, and the residual:

$$r_i = \lambda_{\text{calculated}}(x_i) - \lambda_i$$

In addition, the RMS error of the residuals is calculated:

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N r_i^2}$$

Figures 40 and 41 show the results obtained with both models using the same set of lines.

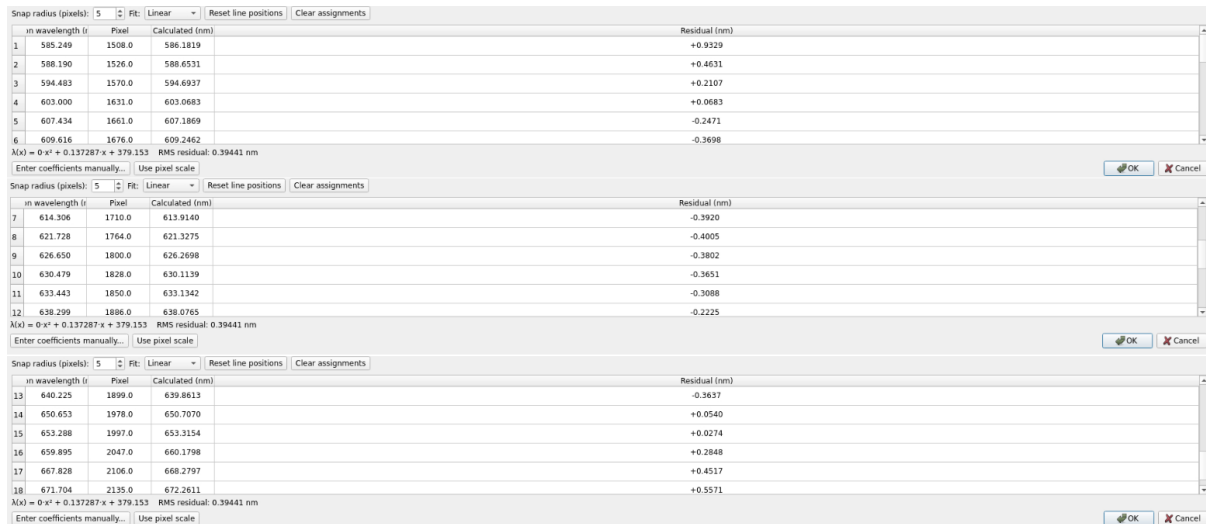


Figure 40: Calibration using a linear fit. For each line, the selected position, calculated wavelength, and corresponding residual are shown.

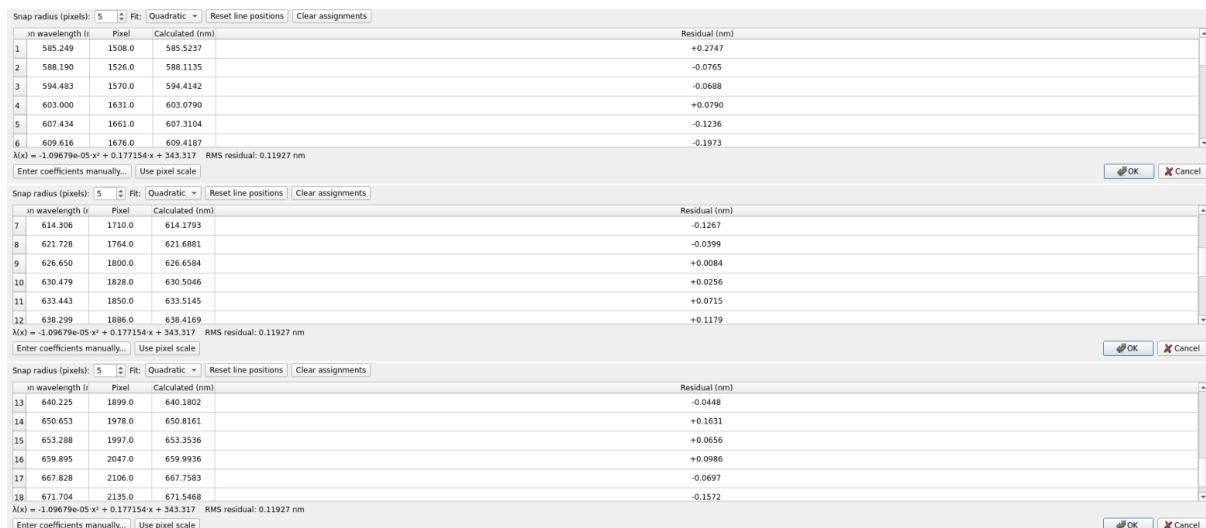


Figure 41: Calibration using a quadratic fit with the same set of spectral lines.

For the set shown, the quadratic fit reduces the RMS residual compared with the linear model, so it was used as an approximation to transform the pixel axis into wavelength.

The obtained coefficients are stored and can be reused in later acquisitions without repeating the calibration procedure.

6.5 Representation and peak detection

Once the calibration is defined, the horizontal axis of the graph can be represented directly in wavelength.

Figure 42 shows the first spectrum of the neon set after applying the quadratic scale and min-max normalization.

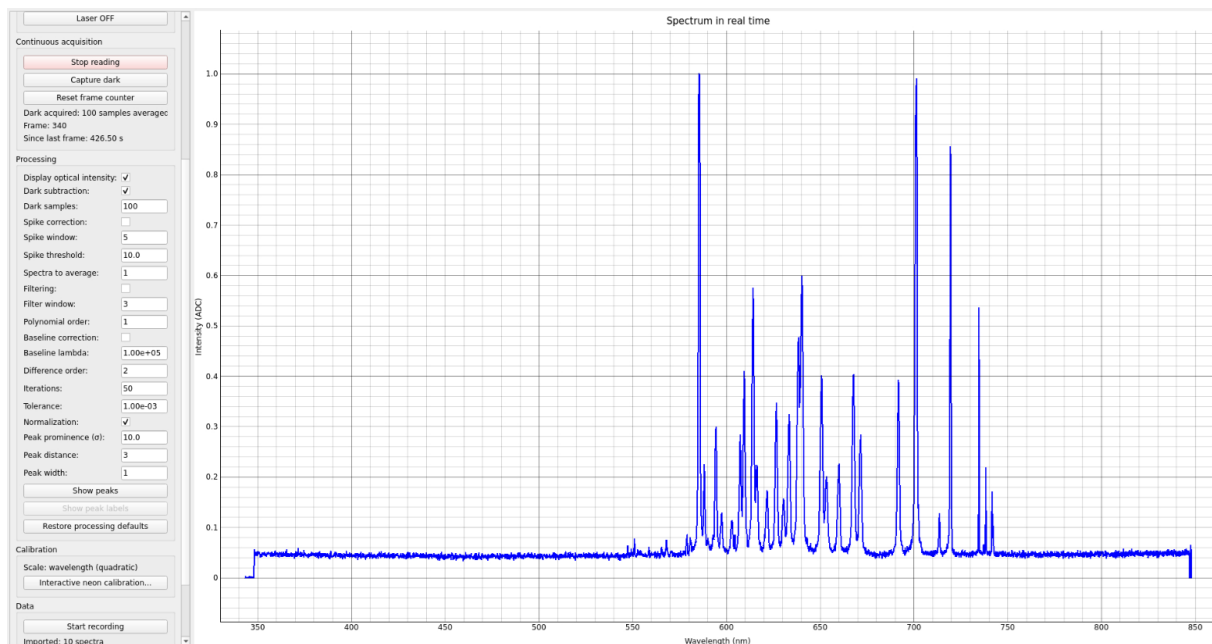


Figure 42: Normalized neon spectrum represented in wavelength using the experimentally obtained quadratic calibration.

The peak-detection tool can be enabled independently of the rest of the processing. The detected maxima are represented by markers and can optionally be labeled with their spectral position and amplitude.

Figure 43 shows an example of automatic detection on a non-normalized neon spectrum.

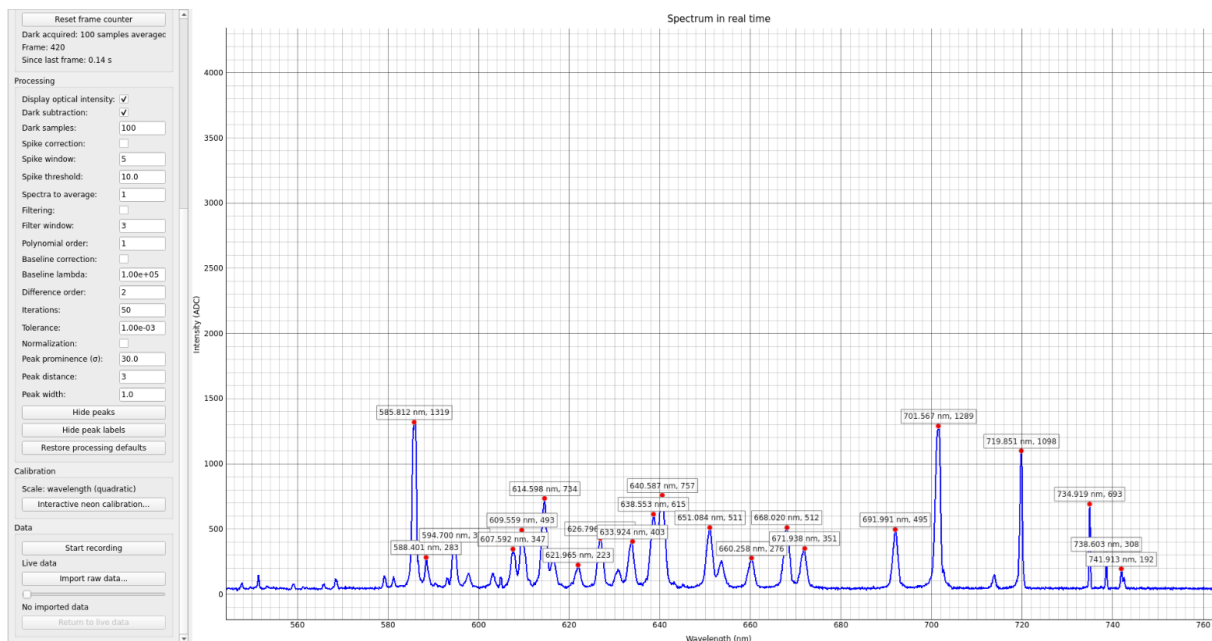


Figure 43: Automatic detection and labeling of maxima in a neon spectrum represented in wavelength.

Peak detection is used as a support tool for locating spectral structures and not as an automatic chemical-identification mechanism.

6.6 Storage and reprocessing

The program allows complete sequences of unprocessed spectra to be stored in compressed NumPy files (.npz). In addition to the original frames, the file can preserve information associated with the acquisition, such as the *dark* reference, spectral calibration, integration time, and processing configuration.

These files can later be imported again from the interface. The stored spectra are browsed using a slider and processed using the configuration active at that time.

This strategy makes it possible to separate experimental acquisition from later analysis: the same measurement can be reprocessed with different smoothing, *baseline* correction, normalization, or peak-detection parameters without modifying the original data.

6.7 Test with an independent spectral source

In addition to the neon lamp, a red LED was used as an additional optical source to verify the response of the system to a different emission spectrum. A new *dark* acquisition was performed for this measurement, maintaining the conditions corresponding to the test.

Figure 44 shows the spectrum obtained by powering the LED with 2.8 V.

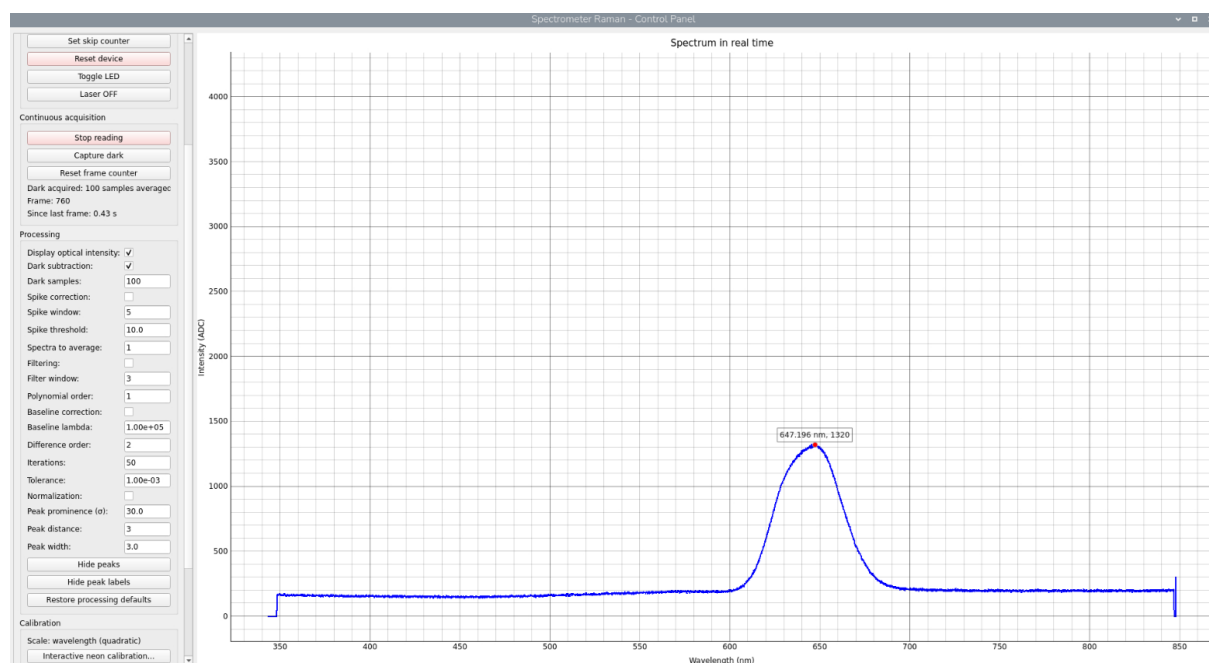


Figure 44: Spectrum measured for a red LED powered at 2.8 V, after performing a new dark-signal acquisition.

Unlike the narrow lines observed with the neon lamp, the LED exhibits a considerably broader emission band. This measurement qualitatively verifies that the system can record spectral distributions with different characteristics and not only discrete lines.

7. Raman measurement results and limitation analysis

Once the system's ability to acquire spectral signals using the neon lamp and other optical sources had been validated, tests aimed at detecting Raman scattering were performed.

Isopropyl alcohol and sugar were used as initial samples because of the expected presence of relatively intense Raman bands and their ease of handling. However, in the configurations tested it was not possible to reproducibly identify bands that could be assigned to the Raman signal of these samples.

This negative result does not, by itself, make it possible to determine which stage of the instrument limited the measurement. For this reason, additional tests were performed and different possible sources of signal loss were analyzed.

7.1 Verification using a commercial spectrometer

In order to determine whether the absence of signal was associated with the developed spectrometer, a Thorlabs CCS200 compact spectrometer was additionally used.

The input fiber of the commercial instrument was coupled to the optical path of the system at a position before the collimating lens of the developed spectrometer. In this way, an important part of the excitation and sample-collection stage was maintained, mainly replacing the dispersive and detection stage.

As a reference, the dark signal and the spectrum of a neon lamp were first recorded.

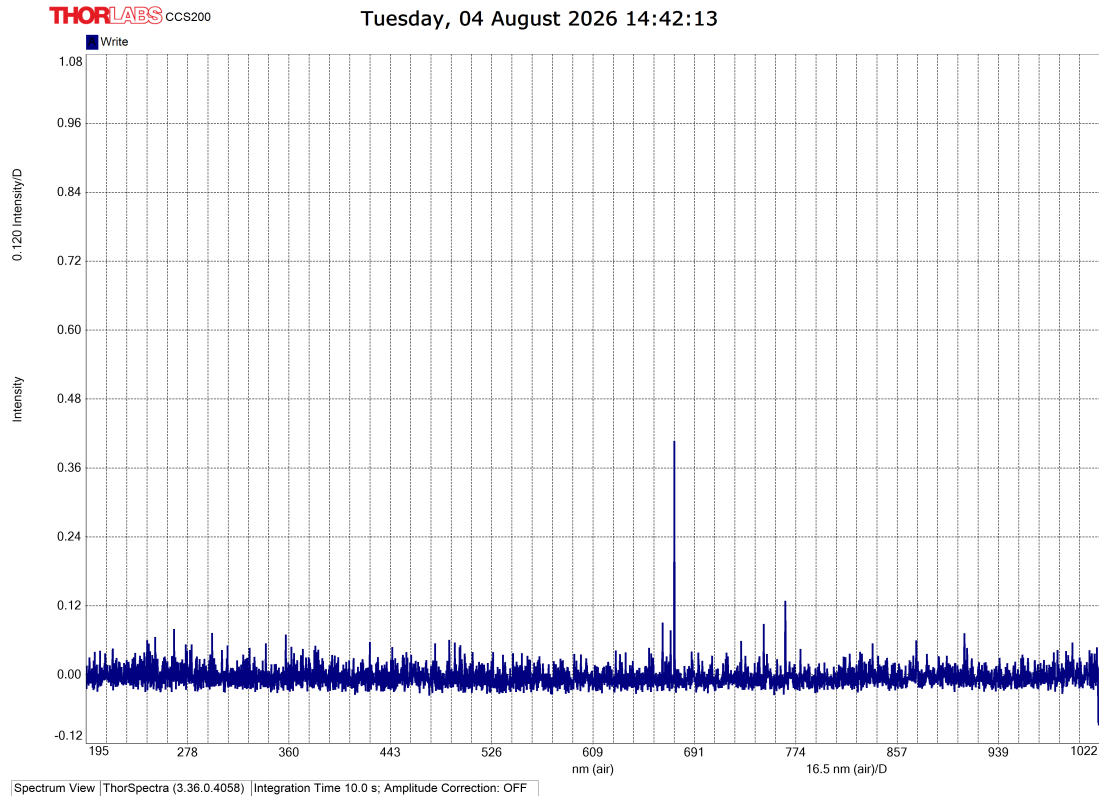


Figure 45: Dark signal recorded using the Thorlabs CCS200 spectrometer ($t_{int} = 1$ s).

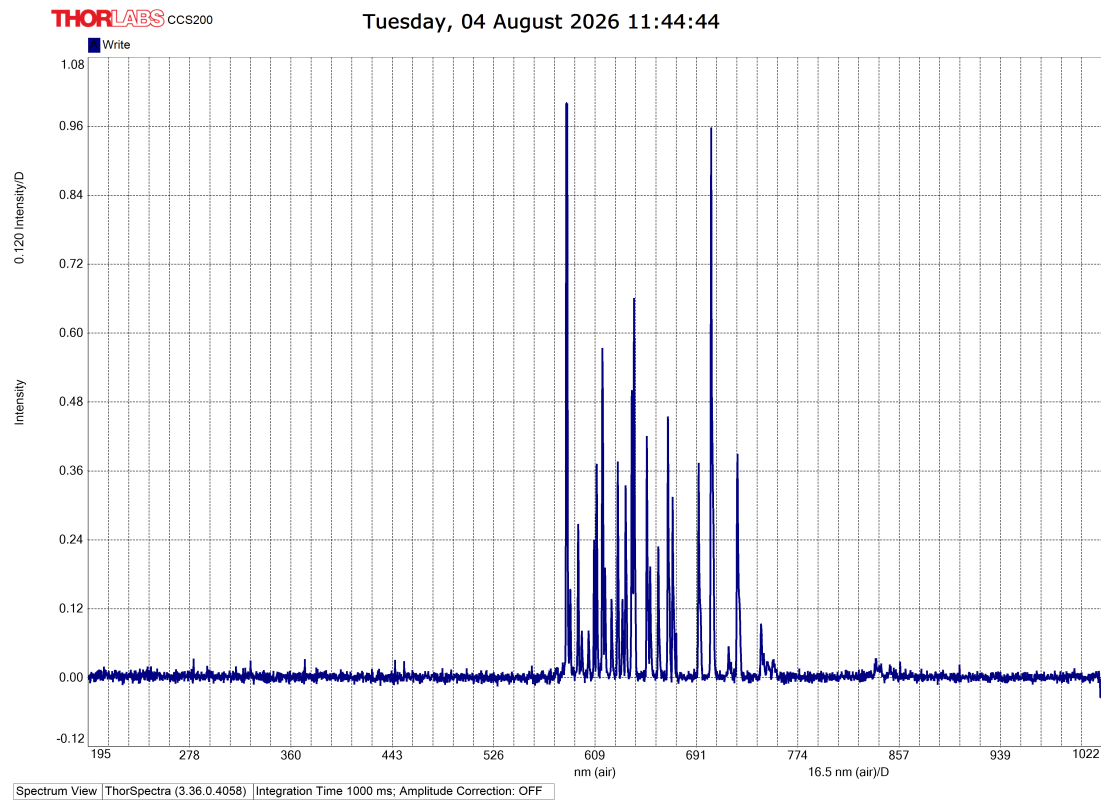


Figure 46: Spectrum of a neon lamp recorded using the Thorlabs CCS200, used to verify instrument operation and optical coupling ($t_{int} = 1$ s).

An isopropyl-alcohol measurement was then performed using an integration time of 10 s. The result is shown in Figure 47.

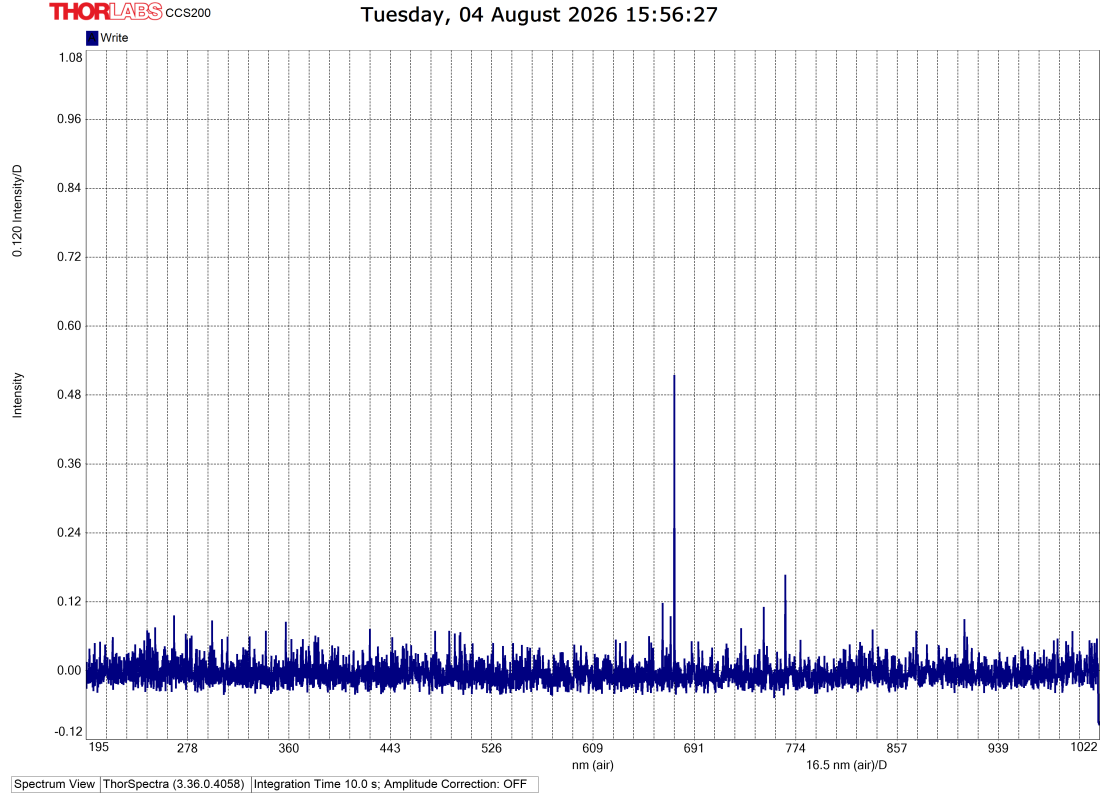


Figure 47: Measurement of isopropyl alcohol using the Thorlabs CCS200 spectrometer. No reproducible Raman bands were identified above the background level ($t_{int} = 10$ s).

The absence of a clear Raman signal also with the commercial spectrometer is consistent with the measurements performed using the developed system. This test suggests that the main limitation is not necessarily located in the linear detector or in the system’s own dispersive stage.

However, this conclusion should be considered only indicative. Coupling into the CCS200 fiber introduces additional alignment, numerical-aperture, and collection-efficiency conditions, so low efficiency at this point could also prevent detection of a weak signal.

Overall, the results shift attention toward the optical stage before the spectrometer, particularly toward the excitation, collection, and transfer efficiency of Raman light from the sample to the input of the system.

7.2 Feasibility of a low-cost Raman spectrometer

The absence of experimentally obtained signal does not imply that the general proposed architecture is unfeasible.

The literature includes developments of low-cost Raman spectrometers built using commercial components, 3D-printed elements, and custom acquisition electronics. For example, Aydogan and Tasal developed a spectrometer with a Crossed Czerny–Turner configuration, linear CCD detector, Nucleo board, and Raspberry Pi, with which they reported the acquisition of Raman spectra of Rhodamine B [3].

Likewise, the open OpenRAMAN project demonstrates the possibility of performing Raman measurements using an excitation source of approximately 4.5 mW, comparable to the power used in this work [7].

Therefore, the nominal laser power or the use of a low-cost detector cannot, in isolation, be considered a sufficient explanation for the absence of signal. The overall efficiency of the optical system and, especially, its alignment are equally relevant factors.

7.3 Optomechanical limitations of the prototype

One of the main differences between the built prototype and reference systems such as OpenRAMAN lies in the optomechanical implementation.

In the developed system, commercial optical components were used in combination with adapters and mounts manufactured by 3D printing. In some cases, the printed parts are used as an interface between components from different manufacturers and, in others, they directly constitute the mount for the optical element.

In addition, due to restrictions in the available components, several mounts were supported using two longitudinal rods instead of four. The diffraction grating and the 45° support for the dichroic mirror were also implemented using printed parts.

This solution made it possible to build and modify the prototype quickly, but introduced several limitations:

- lower mechanical rigidity of the assembly;
- dimensional tolerances associated with 3D-printing manufacture
- difficulty in guaranteeing concentricity between the different optical axes
- absence of fine angular or lateral adjustment mechanisms
- difficulty reproducing a configuration after disassembling or modifying a part
- possibility of small displacements when tightening screws or fasteners

These limitations are especially important in a Raman spectrometer, since the signal of interest is considerably weaker than the excitation radiation.

In fact, the OpenRAMAN documentation identifies alignment as one of the most critical aspects of the design. The system incorporates specific mechanisms to control tilt and decentering, in addition to allowing fine adjustment of the dichroic mirror to correctly position the image on the spectrometer input aperture [8].

Even in commercial mounts, small angular errors and decentering can significantly degrade optical transfer. Therefore, in a system where several positions depend on printing tolerances and manual adjustments, these effects may accumulate along the optical path.

7.4 Hypotheses regarding the absence of Raman signal

Based on the tests performed, it was not possible to identify a single cause responsible for the absence of signal. However, the results make it possible to propose different hypotheses.

7.4.1 Losses due to misalignment

The hypothesis considered most likely is insufficient collection efficiency due to small accumulated alignment errors.

Correctly observing the spectrum of a neon lamp demonstrates that the stage formed by the slit, collimation, grating, image formation, and detector can transmit and resolve a sufficiently intense optical signal. However, this does not guarantee that the weak Raman radiation generated in the sample is focused exactly onto the same aperture.

Unlike an intense calibration source, the available Raman signal is small, so losses that are of little relevance during a neon measurement may become decisive during a Raman measurement.

7.4.2 Collection efficiency at the sample

Another possible limitation corresponds to the sample volume that is effectively illuminated and observed by the system.

To maximize the Raman signal, the region excited by the laser must spatially coincide with the volume from which the collection system forms the image on the slit. A small displacement between both volumes can strongly reduce the number of collected Raman photons even if the laser beam passes correctly through the sample.

This condition is especially difficult to verify in an implementation without micrometric movements or fine angular adjustments.

7.4.3 Coupling to the slit

The aperture used in the system has a width of $50\text{ }\mu\text{m}$. If the image formed by the collection optics is displaced with respect to this aperture, a significant fraction of the light may be lost.

Successful detection of neon made it possible to experimentally adjust the distances of the collimator and the dispersive system. Once an image of acceptable quality was obtained, 1, 5, and 10 mm spacers were manufactured to facilitate reproduction of these distances.

However, these spacers mainly make it possible to reproduce the axial position and do not correct tilt or decentering errors. For this reason, an apparently correct spectral configuration does not necessarily guarantee maximum coupling efficiency from the sample.

7.4.4 Mechanical rigidity and repeatability

The use of rods as longitudinal support made it possible to adapt the instrument during development, but represents an important limitation for a fine-optimization stage.

An architecture based on a rigid optical plate, on which the different subunits are positioned and secured independently, could reduce the propagation of errors between mounts and facilitate local adjustments.

This philosophy is closer to that used in more mature open Raman instruments, where 3D printing can be used for bases, housings, or auxiliary elements, while critical degrees of freedom are controlled through more rigid and reproducible positioning systems.

8. Conclusions and future work

The development carried out made it possible to implement and validate an important part of the instrumental chain required for the construction of a portable and low-cost Raman spectrometer.

At the acquisition level, the control signals required by the TCD1304DG CCD sensor were correctly generated, digital frame acquisition was performed, and the data were transmitted to a Raspberry Pi for visualization and processing. The detector response to different illumination sources was also verified, and a graphical interface capable of configuring acquisition, capturing dark references, applying different processing stages, and storing the acquired data was developed.

From the spectral point of view, the use of a neon lamp made it possible to resolve multiple spectral lines and perform an experimental calibration of the wavelength axis. In addition, the emission of a red LED was correctly detected, observing the system response to a broadband spectral source.

The design of the dedicated electronics for the instrument was also completed. PCB A, which integrates the microcontroller, detector, analog-conditioning, power-supply, and communication stages, was manufactured and tested during development, verifying its operation for data acquisition and transmission. PCB B, intended for motor and Peltier-module control, was designed but could not be experimentally validated within the available time.

In parallel, the mechanical integration of the optical system was designed and implemented using mounts available in the laboratory and parts manufactured by 3D printing. These parts made it possible to adapt components from different manufacturers, position the calibration source and samples, reproduce certain optical distances, and incorporate auxiliary elements such as spacers and diaphragms.

The stage that could not be conclusively validated was the acquisition of a Raman signal from a sample. Tests were performed with different samples and configurations without identifying reproducible Raman bands above the background level.

As an additional verification, the same excitation and collection path was connected to a commercial spectrometer. In this configuration, it was also not possible to identify a clear Raman signal. This result does not make it possible to definitively attribute the absence of signal to a single cause, since fiber coupling and the connection geometry itself introduce additional losses. However, it provides an indication that the main limitation may be located before the dispersion and detection stage, particularly in excitation, collection, and optical-coupling efficiency.

During assembly, different limitations related to mechanical repeatability and control of the optical path were also observed. The combined use of mounts available in the laboratory, 3D-printed adapters, and a rod system made it possible to build and adjust the prototype, but made fine position and angle adjustments difficult. Likewise, in some tests, the presence of stray-light paths outside the intended optical path was observed, which made it necessary to incorporate additional diaphragms.

For this reason, before making major modifications to the acquisition electronics or to the processing algorithms, improving the rigidity, repeatability, and adjustment capability of the optomechanical system is considered a priority.

Although a conclusive Raman signal was not obtained, the prototype made it possible to validate a large part of the subsystems required for this type of instrument and to establish a functional platform on which development can continue. In particular,

acquisition electronics, configurable firmware, a user interface, spectral-calibration and processing tools, and an optical design whose response could be verified using known spectral sources remain available.

8.1 Future work

Based on the results obtained and the limitations encountered during development, different lines of work were identified for a future continuation of the project.

8.1.1 Signal processing

The implemented processing pipeline makes it possible to perform dark correction, spike removal, smoothing, baseline correction, normalization, and peak detection. However, different techniques could be incorporated in future versions:

- Incorporation of dimensionality-reduction methods, such as PCA, particularly for the analysis of large sets of spectra.
- Evaluation of spectral derivatives to improve separation of overlapping bands.
- Implementation of high-pass or band-pass filters when required by the application.
- Evaluation of *spectral stripping* techniques.
- Comparison of other normalization methods, such as normalization by maximum, by spectral area, or using internal or external standards.
- Automatic selection of the Savitzky–Golay filter window based on the expected width of the spectral bands.
- Development of automatic chemical-identification algorithms based on comparison with Raman libraries.
- Implementation of quantitative methods for concentration estimation, accompanied by an appropriate calibration and validation procedure.

8.1.2 Acquisition and processing software

The following improvements could be incorporated into the software developed in Python:

- Direct conversion of the wavelength axis to Raman shift, expressed in cm^{-1} , within the graphical interface.
- Implementation of a physical spectrograph model for calibration, instead of relying exclusively on an empirical polynomial fit.
- Validation of the calibration using a second independent spectral source.
- Explicit storage of additional experimental metadata, such as sample identification, laser power, and optical configuration.
- Validation of the checksum of each frame also on the Python side.

- Automatic comparison of detected peaks with a Raman library.
- Automation of the selection and rejection of calibration lines according to the residual obtained during fitting.
- Greater separation between the graphical-interface, acquisition, and processing modules if the software continues to increase in complexity.

8.1.3 Optical system and Raman acquisition

The main priority for an experimental continuation should be to improve the efficiency and repeatability of the optical chain. The proposed actions include:

- Redesign the assembly on a more rigid optical base.
- Use kinematic or commercial mounts for the most critical components, especially the dichroic mirror and diffraction grating.
- Incorporate a greater number of support points or rods when necessary to increase mechanical rigidity.
- Implement fine position and angular adjustments for the slit, dichroic mirror, grating, and sample.
- Define a systematic alignment procedure using targets or intermediate references.
- Measure optical power stage by stage to characterize the losses introduced by each element.
- Characterize the total collection efficiency from the sample to the detector.
- Incorporate diaphragms, baffles, or internal enclosures to reduce the presence of stray light outside the intended optical path.

8.1.4 Mechanical design and 3D printing

The 3D-printed parts made it possible to adapt the available elements and assemble the prototype. To improve mechanical repeatability, the following modifications could be made:

- Redesign critical parts with more precise mechanical references and greater ease of fine adjustment.
- Reduce possible light paths outside the optical path through geometries that incorporate apertures and optical blocking.
- Replace the most sensitive printed parts with commercial mounts or fine-adjustment mechanisms.
- Evaluate integration on a rigid base instead of relying mainly on the rod-and-adaptor system.

8.1.5 Hardware

Due to time constraints, it was not possible to perform a complete experimental validation of all blocks included in the final hardware.

In particular, PCB B, which incorporates the motor controllers and the power stage for the Peltier module, was not tested during this stage of the project. Its validation therefore remains a pending task, including operation of the drivers, the buck converter, and the thermal-stabilization loop.

The USB interface provided on PCB A was also not ultimately used. During the initial tests, correct operation of this port was not obtained and, since communication through UART made it possible to continue system development, it was decided to postpone its diagnosis. As future work, both the associated hardware and the firmware configuration should be reviewed to determine the origin of the problem.

It would also be advisable to complete the electrical and thermal validation of both boards under the actual operating conditions of the instrument.

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